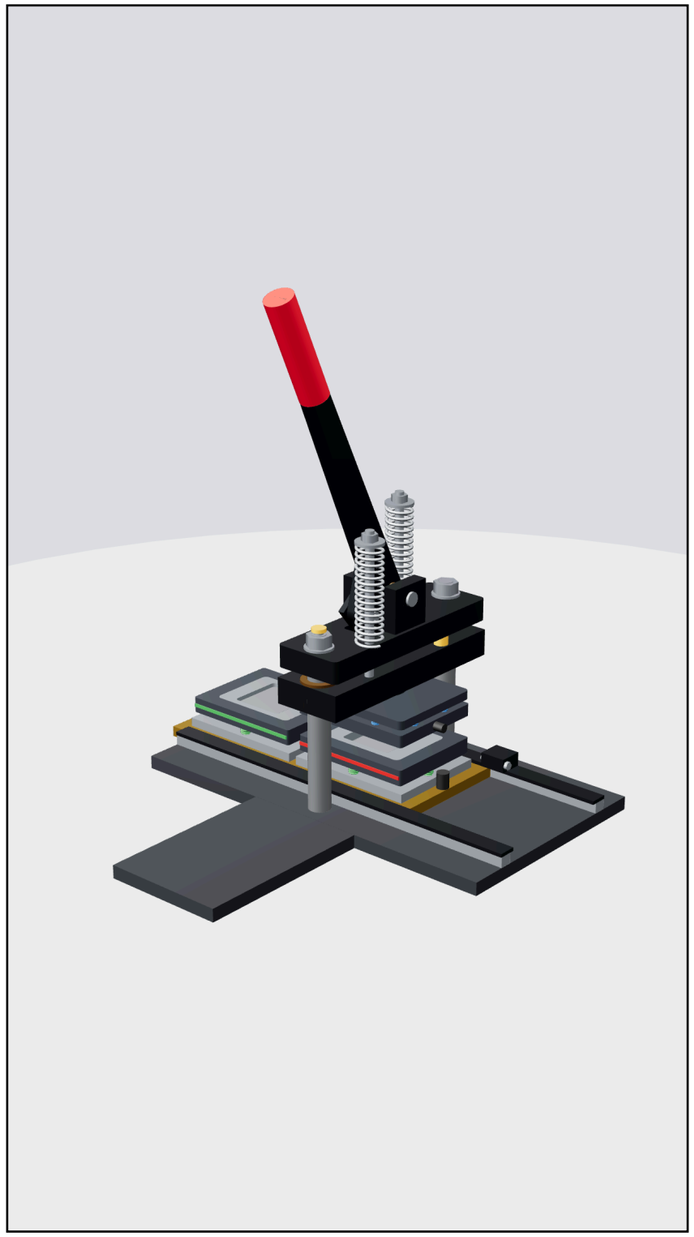
MagPress M4

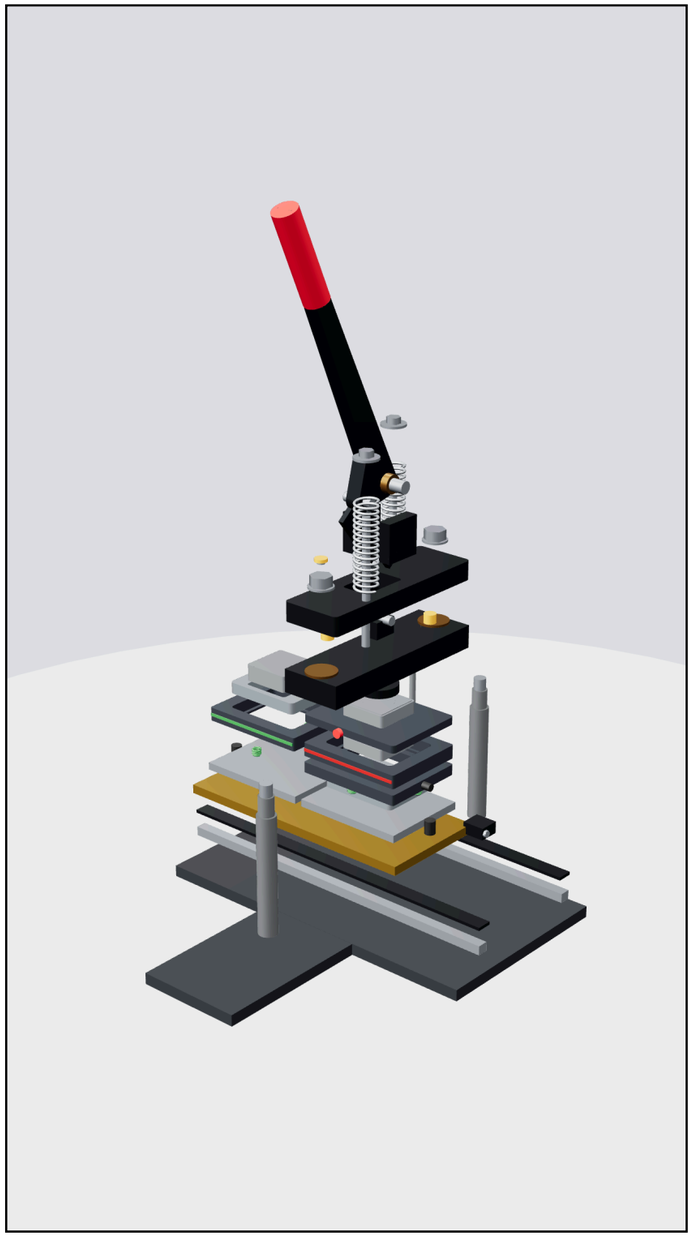
Photo-magnet press · drawing set · Rev C.1 (release)

One universal die + size kits: 50 × 50, 63.5 × 63.5, 53 × 80, 65 × 90 mm

DRAWING LIST

DRG NO.	TITLE	QTY	SHEET
MP-001	GENERAL ARRANGEMENT & HEIGHTS	—	2
MP-101	BASE PLATE	1	3
MP-102	GUIDE RAIL	2 (1 × -F, 1 × -B)	4
MP-103	KEEPER STRIP (GIB)	2 (1 × -F, 1 × -B)	5
MP-104	STATION STOP BLOCK	2	6
MP-105	DIE TABLE (SLIDE)	1	7
MP-106	DIE TABLE STOP PIN	1	8
MP-107	COLUMN	2	9
MP-108	CROWN	1	10
MP-109	RAM	1	11
MP-110	CLEVIS	1	12
MP-111	DIE HOLDER	1	13
MP-112	PIVOT CHEEK	2	14
MP-113	HANDLE LEVER (ONE PIECE)	1	15
MP-114	TOGGLE LINK	2	16
MP-116	PIN A / PIN B (TOGGLE)	2	17
MP-117	PIN O (HANDLE PIVOT)	1	18
MP-118	RETURN ROD	2	19
MP-119	PIVOT SPACER	2	20
MP-120	STATION PILOT	1	21
MP-201	UPPER DIE BODY (UNIVERSAL)	1 per press	22
MP-202	UPPER DIE SHANK	1 per press	23
MP-203	PLUG (SIZE KIT)	1 per size kit (4)	24
MP-204	UPPER RING (SIZE KIT)	1 per size kit (4)	25
MP-205	STATION BASE (UNIVERSAL)	2 per press	26
MP-206	PICKUP PEDESTAL (SIZE KIT)	1 per size kit (4)	27
MP-207	PICKUP RING A (SIZE KIT)	1 per size kit (4)	28
MP-208	CRIMP ANVIL (SIZE KIT)	1 per size kit (4)	29
MP-209	CRIMP RING B (SIZE KIT)	1 per size kit (4)	30
MP-210	UPPER CARRIER (UNIVERSAL)	1 per press	31
MP-211	LOWER CARRIER A / B (UNIVERSAL)	A + B per press	32
MP-212	PLUG STUD (SIZE KIT)	1 per size kit (4)	33





Reference renderings (Rev C.1): assembled, handle up (left); exploded (right). Not for measurement.

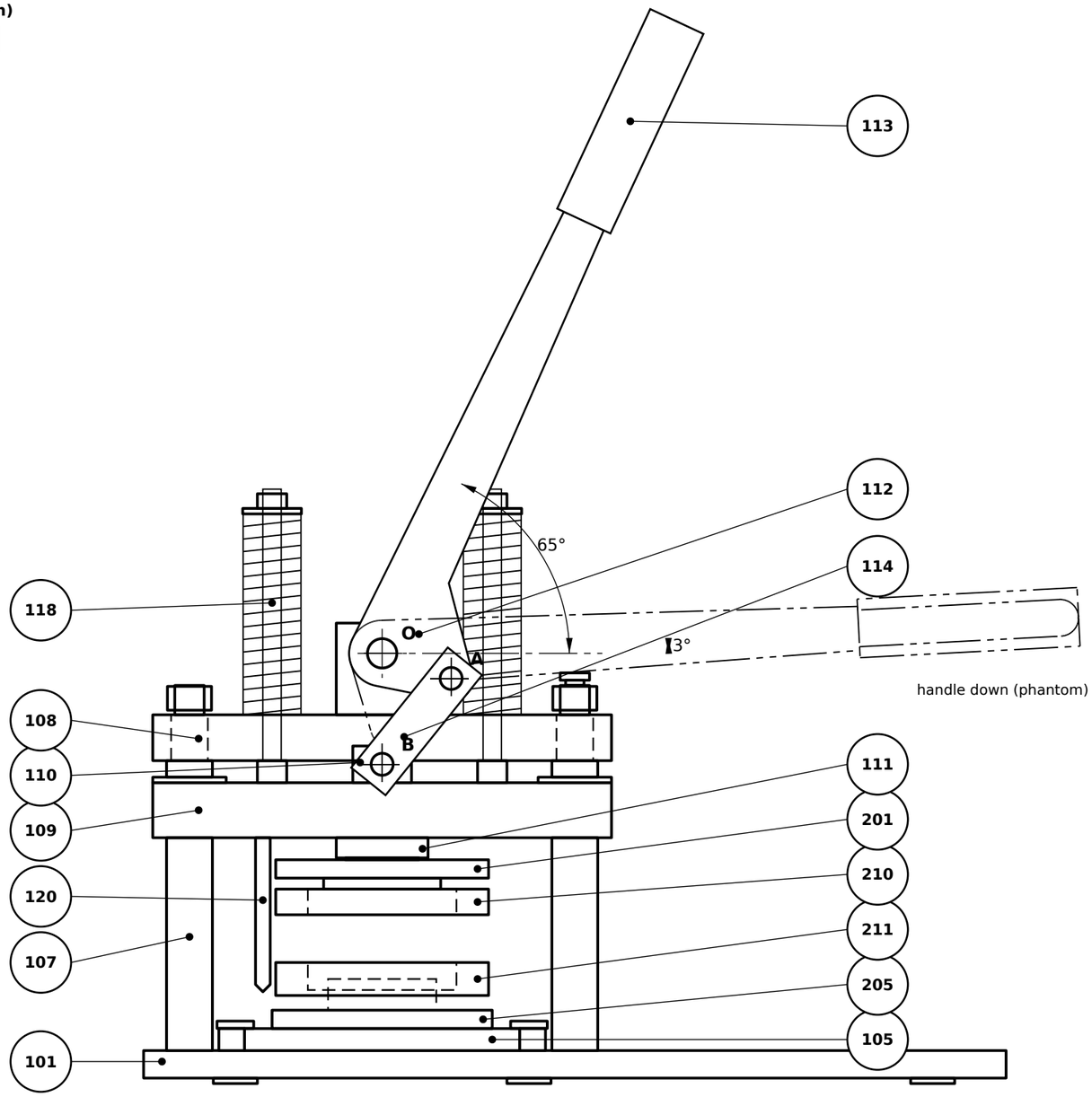
NOTES

1. Units mm. First-angle projection (ISO 128-30); every view is also labelled by name.
2. General tolerances ISO 2768-mK unless stated. Fits ISO 286. Threads ISO metric coarse 6H/6g unless stated. Ra 3.2 unless stated.
3. Column bores in base MP-101, crown MP-108 and ram MP-109 at 210.00 ±0.01: same datum scheme and CNC program.
4. Rev C die: universal parts (MP-201, 202, 205, 210, 211) are made once per press. Size-kit parts (MP-203, 204, 206 to 209, 212) are tabulated for 5050, 6363, 5380, 6590.
5. DXF profiles are supplied for cutting and wire-EDM. If a DXF and a drawing disagree, the drawing governs. Bought parts are in the build pack BOM.
6. Rev B: failure-mode review (lever, pin O, bushings, return springs, pilot, base, 16 kN). Rev C (2026-10-27): universal die + 6-part size kits changed by hand in about a minute; crimp anvil with roll channel; die springs re-rated to ISO 10243 limits. Rev C.1 (2026-10-27), production release: fit fixes from the 3D interference check: ram link pockets (MP-109), crown reliefs for pins A and B (MP-108), push knobs on the table top (MP-105), flush column clamp (MP-101, MP-107), handle stop without jam nut, stop screw M6 × 0.75 × 40 (MP-104).
7. Prove the die fits on a 7075 aluminium set before cutting tool steel.

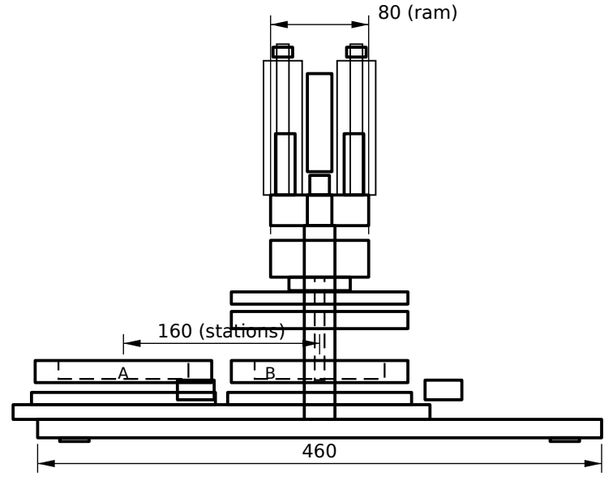
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 1 / 33	
TITLE: **DRAWING SET — COVER & LIST**				
DRAWING NO.: **MP-000**	QTY: —	SCALE: —	UNITS: mm	
MATERIAL: —				
HEAT TREATMENT / FINISH: —				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:

KEY HEIGHTS, handle up (mm from bench)

HEIGHT	LEVEL
586	grip top, handle up
324	return-rod top
234.5	pivot O
201	crown top
176	crown bottom
164	ram top
134	ram bottom
92	upper die bottom
66	lower die top
40	station base top
30	die table top
18	base top
3	base underside (3 mm pads)
0	bench



SIDE ELEVATION (viewed from the left; front →)

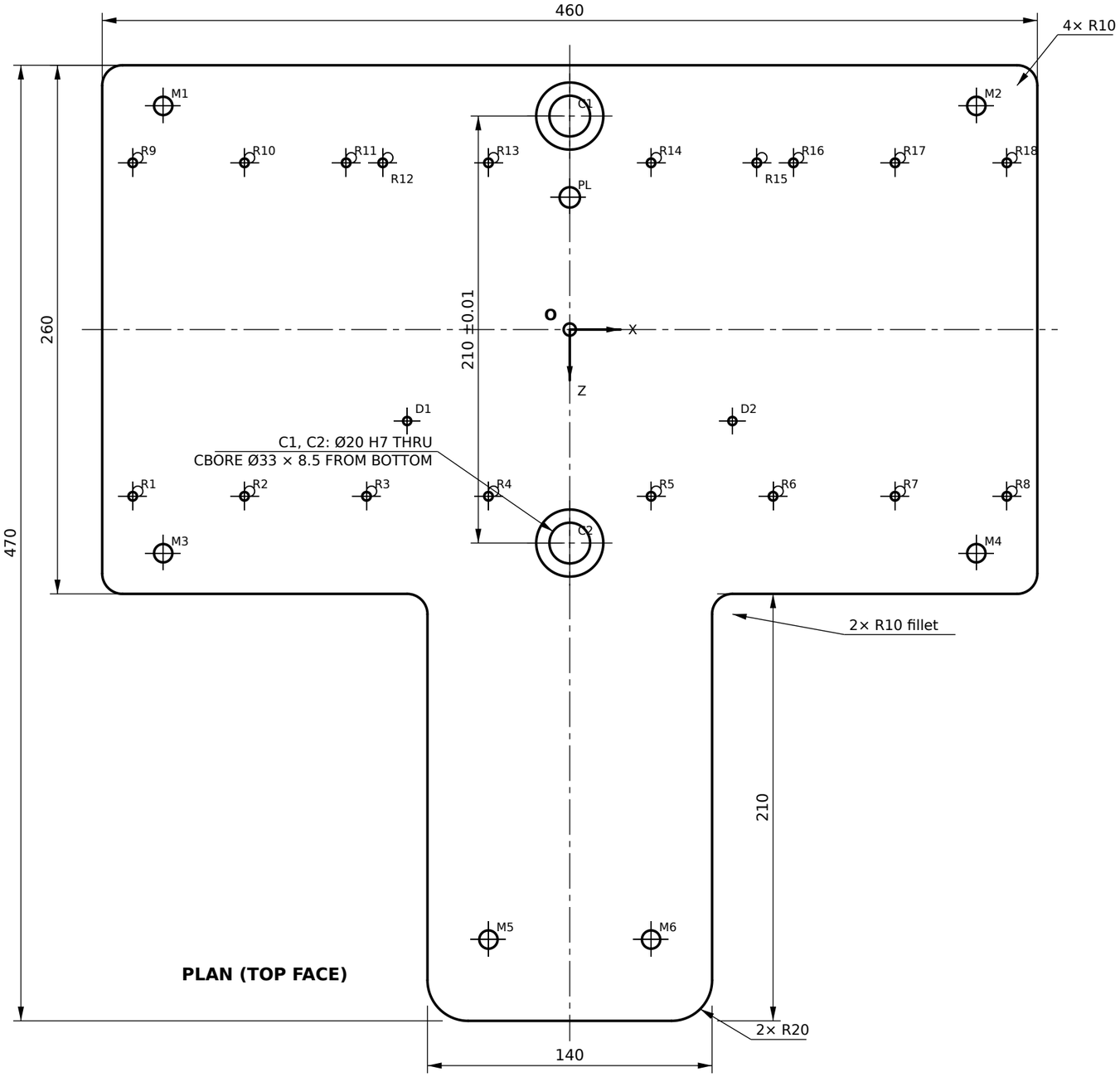


FRONT ELEVATION 1:5 (handle cut off)

NOTES

1. Shown with the handle UP (open), die table at "B under ram", size kit 6363 in the universal die. Heights from the bench top.
2. Toggle: crank O→A 40, link A→B 60, grip centre 320 from O (Rev A 290). Handle 65° → 3° above horizontal. Force gain 6.6 at the start, 34.6 at the end of the stroke.
3. Ram stroke 38.9. Dies meet after 26.0; working travel 12.9. Pickup stroke stops at 35.5. Station pilot MP-120 enters the table at 20.0, before the dies meet.
4. Open stop: 2 urethane bumpers on the ram top. Closed stop: adjustable M10 × 1 screw + urethane cap on the front column top, set so the lever rests at 3°.
5. Rev B: one-piece lever (MP-115 deleted); pin O Ø16; dry bushings; return springs 80 / 245 N; station pilot; 15 mm base. Rev C: universal die + size kits (MP-201 to 212). Rev C.1: seven fit fixes from the 3D check (MP-101, 104, 105, 107, 108, 109).
6. Balloons = drawing numbers (MP-xxx). Pins MP-116 at A and B, MP-117 at O; spacers MP-119 on pin O; stop blocks MP-104 behind the table.

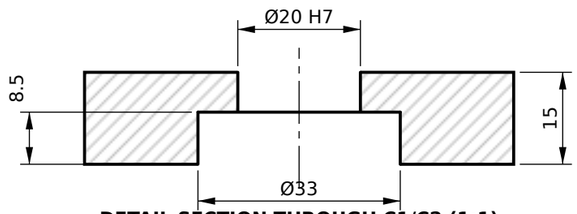
PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **C.1**	SHEET: 2 / 33	
TITLE: **GENERAL ARRANGEMENT & HEIGHTS**			
DRAWING NO.: **MP-001**	QTY: —	SCALE: 1:3 / 1:5	UNITS: mm
MATERIAL: —			
HEAT TREATMENT / FINISH: —			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED



PLAN (TOP FACE)

HOLE TABLE

HOLE	X	Z	DESCRIPTION
C1	0	-105	Ø20 H7 THRU; CBORE Ø33 × 8.5 BOTTOM
C2	0	105	Ø20 H7 THRU; CBORE Ø33 × 8.5 BOTTOM
R1	-215	82	M5 THRU (rail screws)
R2	-160	82	M5 THRU (rail screws)
R3	-100	82	M5 THRU (rail screws)
R4	-40	82	M5 THRU (rail screws)
R5	40	82	M5 THRU (rail screws)
R6	100	82	M5 THRU (rail screws)
R7	160	82	M5 THRU (rail screws)
R8	215	82	M5 THRU (rail screws)
R9	-215	-82	M5 THRU (rail screws)
R10	-160	-82	M5 THRU (rail screws)
R11	-110	-82	M5 THRU (rail screws)
R12	-92	-82	M5 THRU (rail screws)
R13	-40	-82	M5 THRU (rail screws)
R14	40	-82	M5 THRU (rail screws)
R15	92	-82	M5 THRU (rail screws)
R16	110	-82	M5 THRU (rail screws)
R17	160	-82	M5 THRU (rail screws)
R18	215	-82	M5 THRU (rail screws)
M1	-200	-110	Ø9 THRU (bench mounting, M8)
M2	200	-110	Ø9 THRU (bench mounting, M8)
M3	-200	110	Ø9 THRU (bench mounting, M8)
M4	200	110	Ø9 THRU (bench mounting, M8)
M5	-40	300	Ø9 THRU (bench mounting, M8)
M6	40	300	Ø9 THRU (bench mounting, M8)
D1	-80	45	Ø4 × 90° spot, 1.5 deep (detent)
D2	80	45	Ø4 × 90° spot, 1.5 deep (detent)
PL	0	-65	Ø10 × 8 deep (pilot clearance)



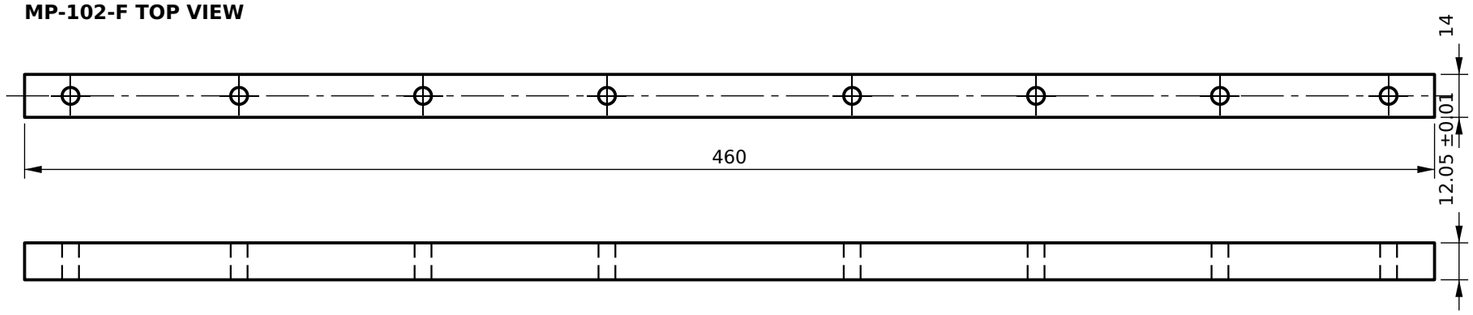
DETAIL SECTION THROUGH C1/C2 (1:1)

NOTES

1. Rev B: 15 mm S355 (was 12 S235). At the 16 kN design load: 112 MPa, 0.26 mm (Rev A: 175 MPa, 0.51 mm). 0.08 mm at a normal 5 kN stroke.
2. Face both sides to 15.0 ±0.1. Flatness 0.05 over the slide area X ±230 × Z ±75, Ra 1.6.
3. C1/C2 Ø20 H7 at 210.00 ±0.01: same datum and program as crown MP-108 and ram MP-109. CBORE Ø33 × 8.5 from the BOTTOM (Rev C.1, was 4.5): the M10 × 16 DIN 7984 clamp screw and Ø32 × 2.5 washer sit flush with the underside.
4. M1–M6: Ø9 THRU bench-mounting holes (M8). Underside: five 3 mm adhesive rubber pads Ø24 at (±200, ±80) and (0, 300).
5. PL: Ø10 × 8 deep clearance for the station pilot MP-120. D1/D2: Ø4 × 90° spots, 1.5 deep (table detents).
6. Hole positions from datum O (midpoint of C1–C2). X to the right; Z toward the front (tongue).
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 3 / 33
TITLE: **BASE PLATE**			
DRAWING NO.: **MP-101**	QTY: 1	SCALE: 1:2.5	UNITS: mm
MATERIAL: S355J2 steel plate 20 mm, profile cut, faced to 15.0			
HEAT TREATMENT / FINISH: Slide area: QPQ nitrocarburize (prototype: black oxide + oil). Rest: black powder coat, bores masked			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

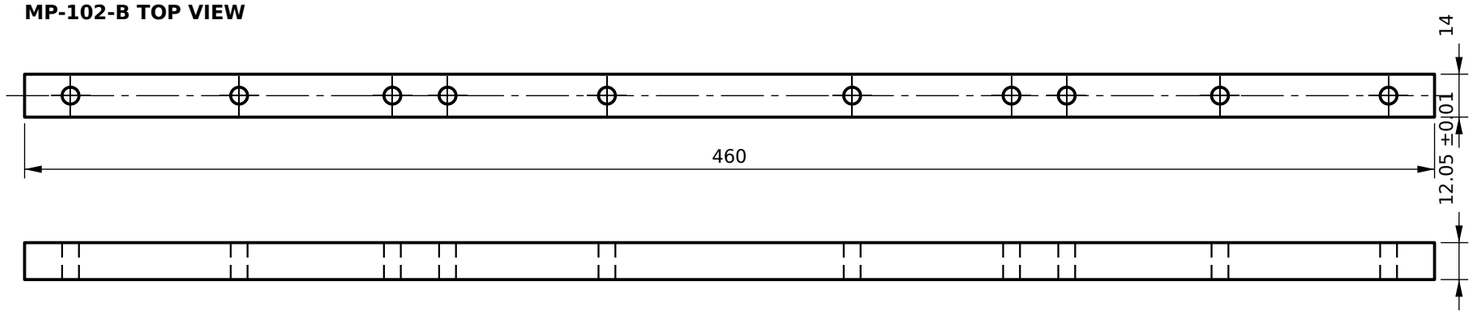
MP-102-F TOP VIEW



Hole X positions (from rail centre), all Ø5.5 THRU:

-215, -160, -100, -40, 40, 100, 160, 215

MP-102-B TOP VIEW



Hole X positions (from rail centre), all Ø5.5 THRU:

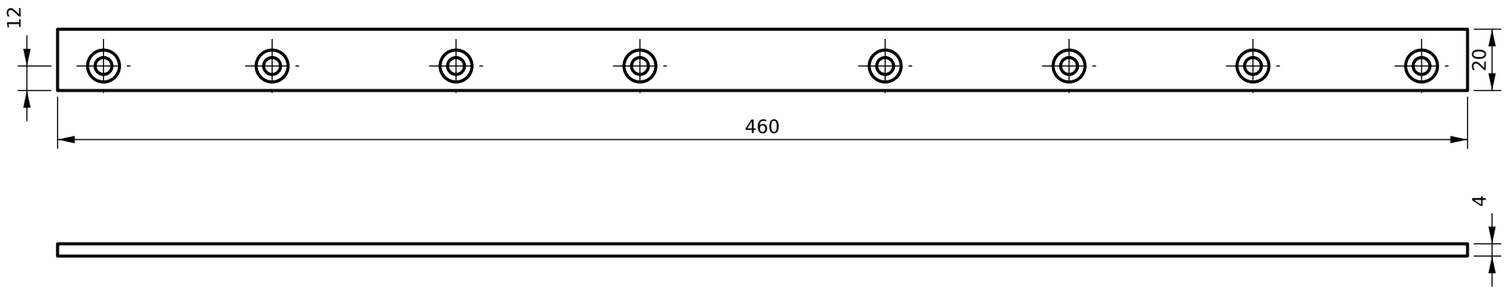
-215, -160, -110, -92, -40, 40, 92, 110, 160, 215

NOTES

1. Height 12.05 ±0.01: equal to die table MP-105 thickness + 0.05 running clearance. Grind both rails together.
2. Ground faces: top, bottom and inner face (faces the die table). Inner face straight within 0.02 over 460.
3. Variant -F: front rail, 8 holes. Variant -B: back rail, 10 holes (includes stop-block screws).
4. Holes Ø5.5 THRU on the rail centreline; hole X positions measured from the rail centre.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 4 / 33
TITLE: **GUIDE RAIL**			
DRAWING NO.: **MP-102**	QTY: 2 (1 × -F, 1 × -B)	SCALE: 1:2	UNITS: mm
MATERIAL: C45 bright flat 16 × 14, ground to 14 × 12.05			
HEAT TREATMENT / FINISH: Black oxide; ground faces oiled			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

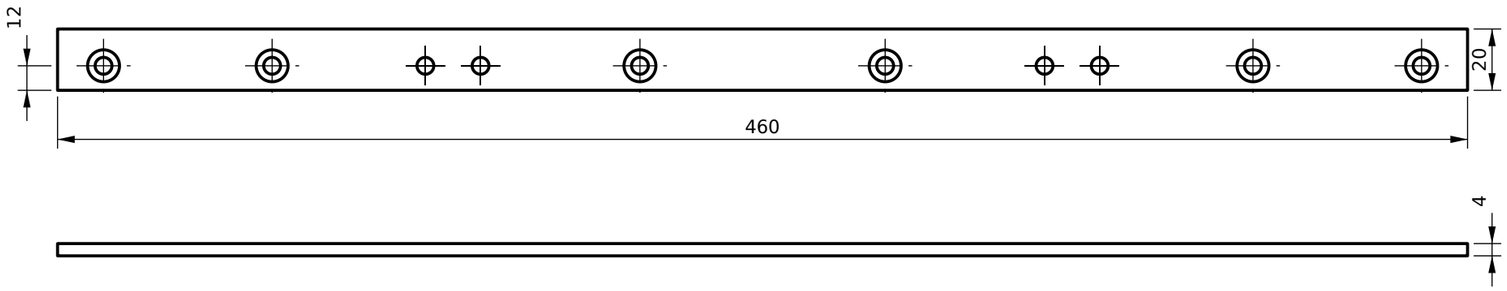
MP-103-F TOP VIEW (inner edge at bottom)



Countersunk (M5 ISO 10642):

-215, -160, -100, -40, 40, 100, 160, 215

MP-103-B TOP VIEW (inner edge at bottom)



Countersunk (M5 ISO 10642):

-215, -160, -40, 40, 160, 215

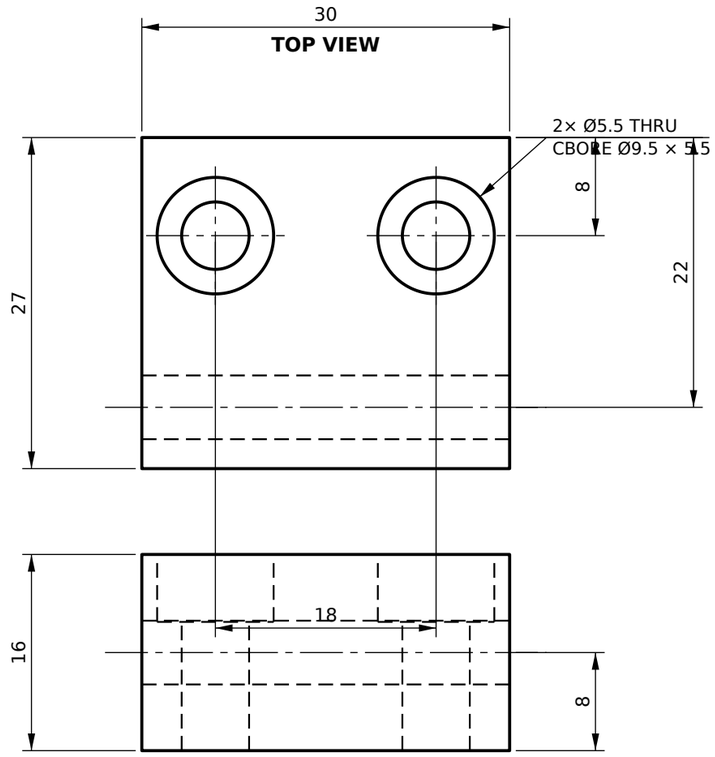
Plain Ø5.5 THRU:

-110, -92, 92, 110

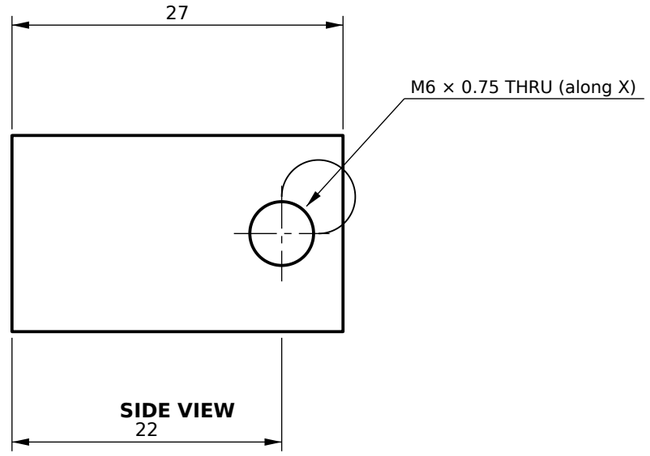
NOTES

1. Sits on top of the guide rail and overlaps the die table edge by 5 mm (inner edge).
2. Countersunk holes: Ø5.5 THRU, 90° countersink to Ø10.4 (for M5 ISO 10642), from top.
3. Variant -B: four plain Ø5.5 holes at X ±92 and ±110 for the stop-block screws (no countersink).
4. Hole line 12.0 from the inner edge. Hole X positions measured from the strip centre.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 5 / 33
TITLE: **KEEPER STRIP (GIB)**			
DRAWING NO.: **MP-103**	QTY: 2 (1 × -F, 1 × -B)	SCALE: 1:2	UNITS: mm
MATERIAL: C45 bright flat 20 × 4			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:



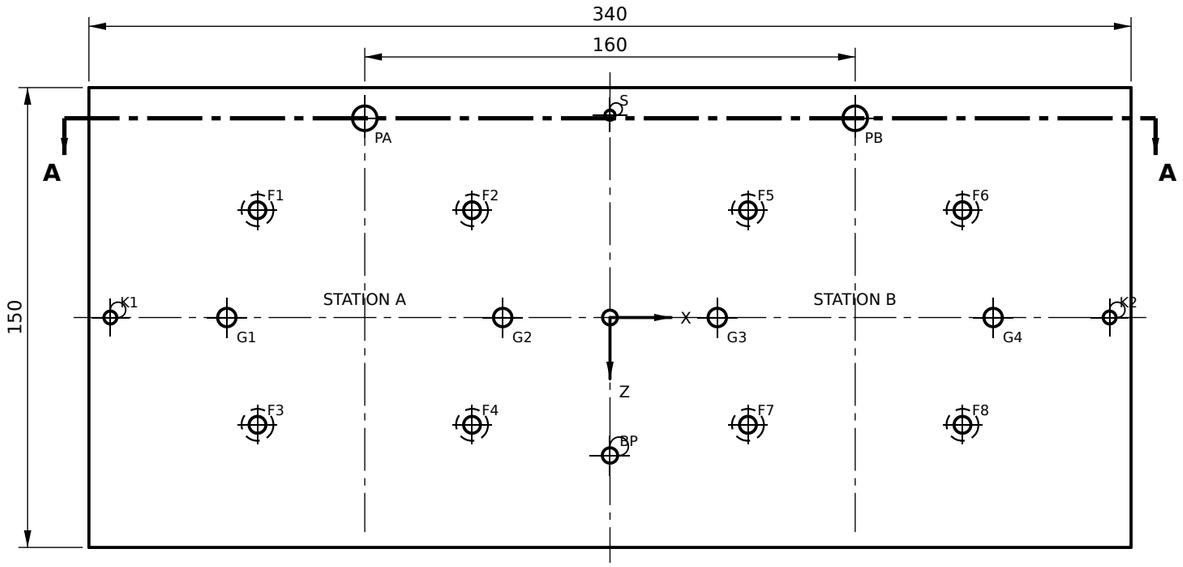
FRONT VIEW (from the die table side)



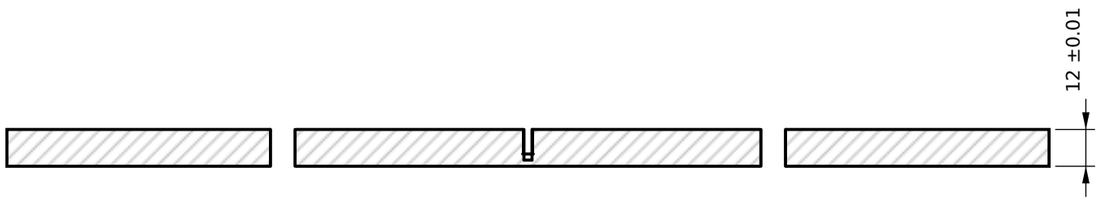
NOTES

1. Bolted through keeper MP-103-B and rail MP-102-B into the base with 2 × M5 × 35 ISO 4762 + threadlocker.
2. Stop screw: M6 × 0.75 × 40 hex screw + thin lock nut (bought; Rev C.1, 25 could not reach through the block). Coarse stop only: the pilot MP-120 sets the exact station position.
3. The 7 mm overhang (Z 20–27) sits above the die table; the table stop pin MP-106 hits the screw tip.
4. Z measured from the back face (the face over the keeper).
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 6 / 33
TITLE: **STATION STOP BLOCK**			
DRAWING NO.: **MP-104**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: C45 bright 30 × 16 flat			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:



PLAN



SECTION A–A (through the pilot holes)

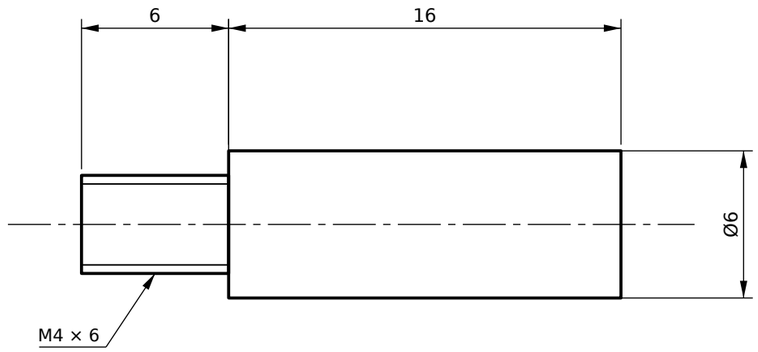
HOLE TABLE

HOLE	X	Z	DESCRIPTION
F1	-115	-35	Ø5.5 THRU, CSK Ø10.4 × 90° BOTTOM (base screw)
F2	-45	-35	Ø5.5 THRU, CSK Ø10.4 × 90° BOTTOM (base screw)
F3	-115	35	Ø5.5 THRU, CSK Ø10.4 × 90° BOTTOM (base screw)
F4	-45	35	Ø5.5 THRU, CSK Ø10.4 × 90° BOTTOM (base screw)
F5	45	-35	Ø5.5 THRU, CSK Ø10.4 × 90° BOTTOM (base screw)
F6	115	-35	Ø5.5 THRU, CSK Ø10.4 × 90° BOTTOM (base screw)
F7	45	35	Ø5.5 THRU, CSK Ø10.4 × 90° BOTTOM (base screw)
F8	115	35	Ø5.5 THRU, CSK Ø10.4 × 90° BOTTOM (base screw)
G1	-125	0	Ø6 H7 × 10 deep (base dowel)
G2	-35	0	Ø6 H7 × 10 deep (base dowel)
G3	35	0	Ø6 H7 × 10 deep (base dowel)
G4	125	0	Ø6 H7 × 10 deep (base dowel)
S	0	-66	M4 × 8 deep (stop pin MP-106)
BP	0	45	M6 THRU (ball plunger)
PA	-80	-65	Ø8 H8 THRU (station pilot), ±0.02
PB	80	-65	Ø8 H8 THRU (station pilot), ±0.02
K1	-163	0	M5 × 8 deep (push knob Ø14, top face)
K2	163	0	M5 × 8 deep (push knob Ø14, top face)

NOTES

1. Thickness 12.00 ±0.01 and flat within 0.03 after anodising. Guide rails MP-102 are 12.05.
2. PA/PB Ø8 H8 THRU take the station pilot MP-120, ±0.02 to the base dowel holes G. Rev C: the station bases MP-205 are fixed for good (F, G); no thumb screws.
3. F1–F8: Ø5.5 THRU, 90° countersink Ø10.4 from the BOTTOM (M5 × 16 ISO 10642, flush). G1–G4: Ø6 H7 × 10 (base dowels).
4. S: M4 × 8 (stop pin MP-106). BP: M6 THRU (ball plunger). K1/K2: M5 × 8 in the top face (Ø14 push knobs; Rev C.1, the Rev C Ø24 end-face knobs reached below the table into the base).
5. Hole positions from the table centre. Station centres at X = −80 (A) and +80 (B), Z = 0.
6. Long edges: 0.5 × 45° chamfer only (they run against the rails).
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **C.1**	SHEET: 7 / 33	
TITLE: **DIE TABLE (SLIDE)**			
DRAWING NO.: **MP-105**	QTY: 1	SCALE: 1:2	UNITS: mm
MATERIAL: EN AW-6061-T6 plate 15 mm, faced to 12.00			
HEAT TREATMENT / FINISH: Hard anodise 25 µm, gold, PTFE-sealed			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

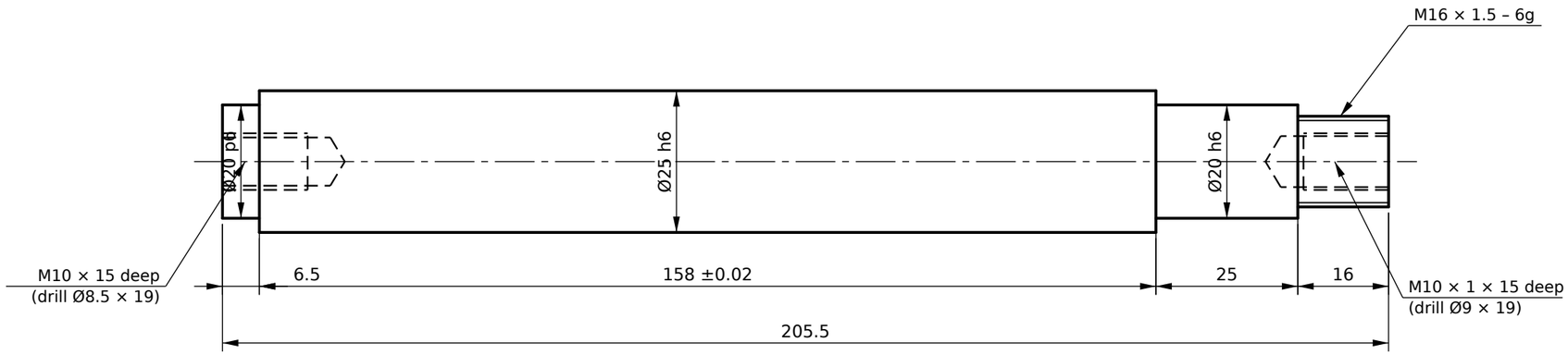


SIDE VIEW

NOTES

1. Screwed into the die table (MP-105, hole S) with Loctite 243.
2. Contact face Ø6 must be square to the axis within 0.05.
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 8 / 33
TITLE: **DIE TABLE STOP PIN**			
DRAWING NO.: **MP-106**	QTY: 1	SCALE: 4:1	UNITS: mm
MATERIAL: C45 Ø8 bright bar			
HEAT TREATMENT / FINISH: Harden 45–50 HRC; black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

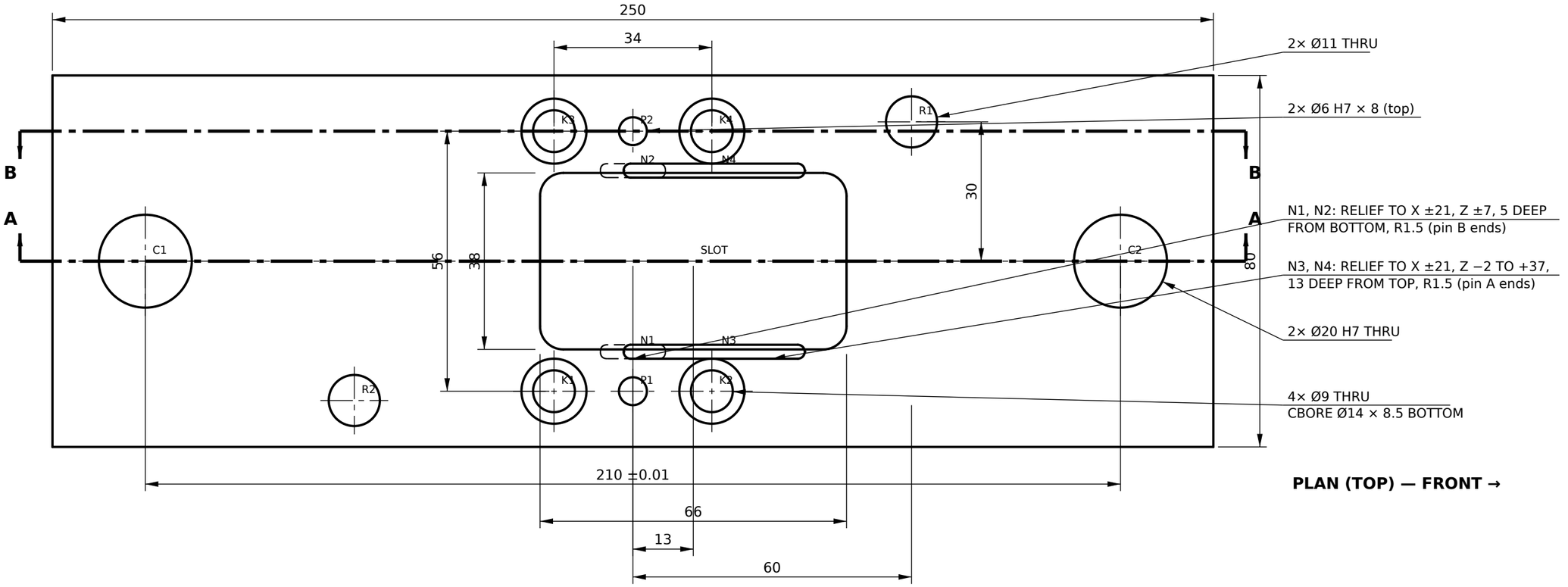


SIDE VIEW

NOTES

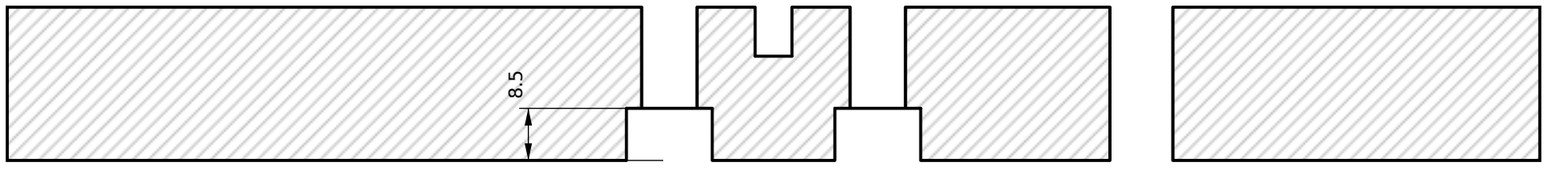
1. Ø25 h6 length 158.00 ±0.02; the two columns matched within 0.01 (sets crown parallelism).
2. Rev C.1: bottom spigot Ø20 p6 × 6.5 (was 10.5), so the base counterbore can be 8.5 deep. Clamp from below: M10 × 16 DIN 7984 + hardened washer Ø32 × 2.5, 40 N·m; head flush with the base underside.
3. Ø20 h6 top spigot passes through the crown (Ø20 H7). Nut M16 × 1.5 + Nord-Lock washer on top, 100 N·m.
4. Rev B: top end M10 × 1 × 15 (both columns). Front column takes the adjustable handle stop: M10 × 1 × 12 set screw with nylon patch + urethane cap (Rev C.1: no jam nut, it had no room).
5. Shoulders square to the axis within 0.01. Run-out of all diameters 0.01 to the Ø25 axis.
6. Chamfers: ends 1 × 45°; steps 0.5 × 45°. Thread undercut DIN 76-A at the M16 thread end.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 9 / 33
TITLE: **COLUMN**			
DRAWING NO.: **MP-107**	QTY: 2	SCALE: 1:1	UNITS: mm
MATERIAL: 100Cr6 (52100 / GCr15) Ø30 bar			
HEAT TREATMENT / FINISH: Ø25 section: induction harden 58–62 HRC, hard chrome 8–12 µm, ground to h6, Ra 0.2			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED





SECTION A–A

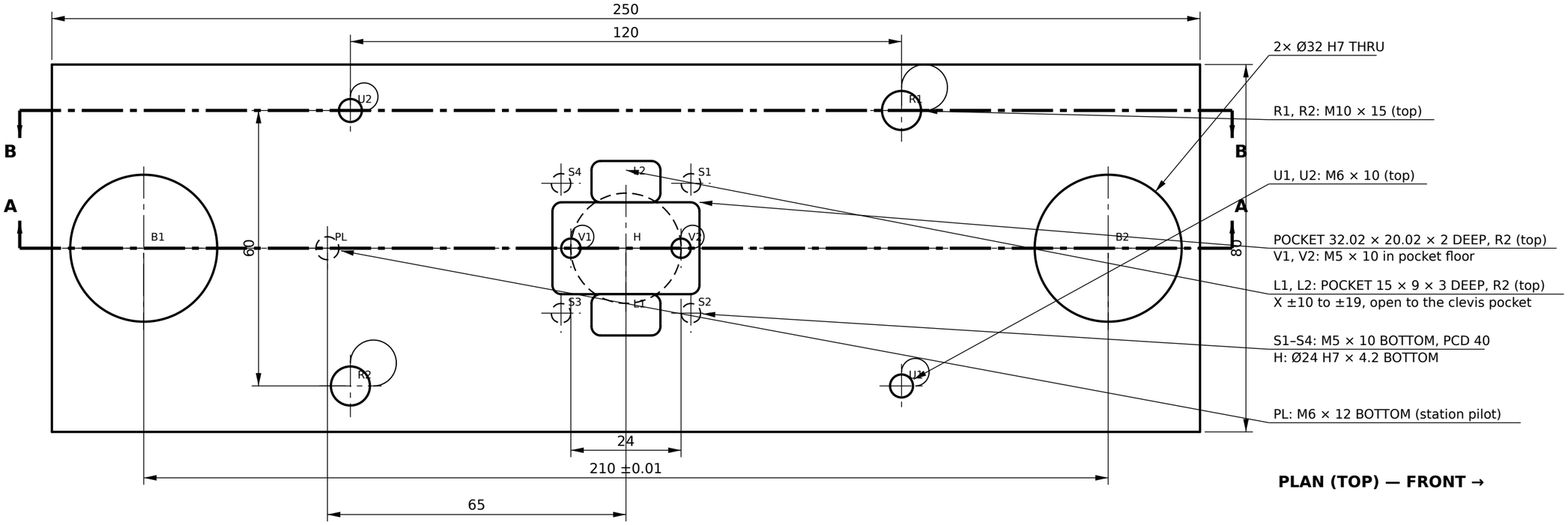


SECTION B–B (X = +28)

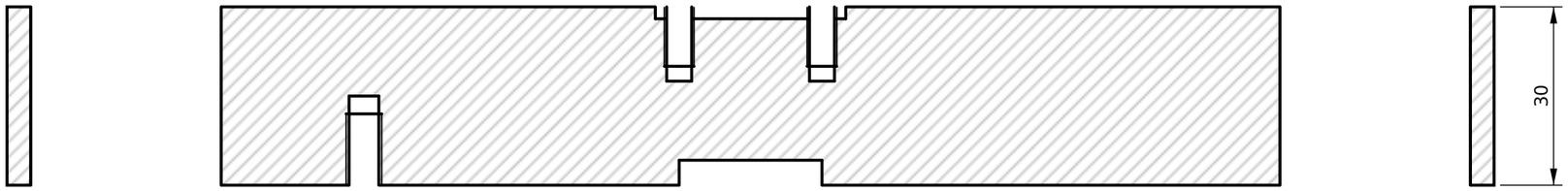
NOTES

1. C1/C2 Ø20 H7 at 210.00 ±0.01: same datum and program as base MP-101 and ram MP-109.
2. Rev B: return-rod holes R1/R2 moved to X = ±30. Design load raised to 16 kN: 192 MPa (C45 N, Re ≈ 340), 0.27 mm.
3. K1–K4: Ø9 THRU, CBORE Ø14 × 8.5 from the BOTTOM (M8 × 30 ISO 4762 + Nord-Lock into the cheeks MP-112).
4. P1/P2: Ø6 H7 × 8 from the top (cheek dowels). Slot 38 × 66, R5, THRU (links pass through).
5. Rev C.1: reliefs in the slot walls, all to X ±21, R1.5. N1/N2 from the BOTTOM, Z −7 to +7, 5 deep: ends of pin B, handle up. N3/N4 from the TOP, Z −2 to +37, 13 deep: ends of pin A, handle down. 212 MPa at 16 kN at the centre section.
6. Top and bottom faces parallel 0.02, flat 0.02. Plan drawn with the front of the press to the right.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

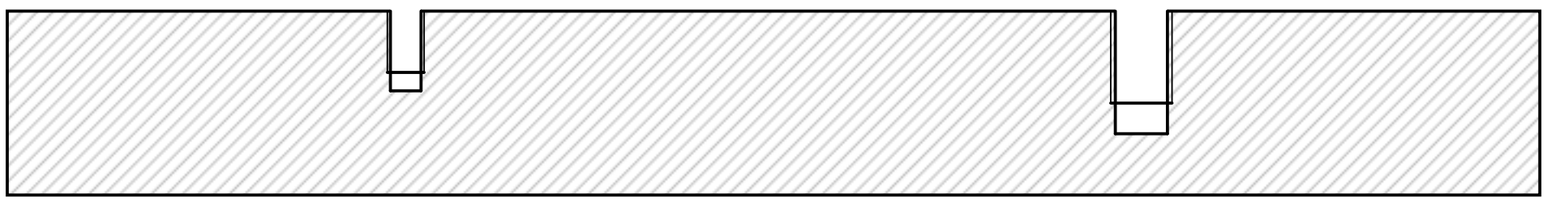
PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 10 / 33
TITLE: **CROWN**			
DRAWING NO.: **MP-108**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: C45 flat 80 × 30, faced to 25			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:



PLAN (TOP) — FRONT →



SECTION A–A

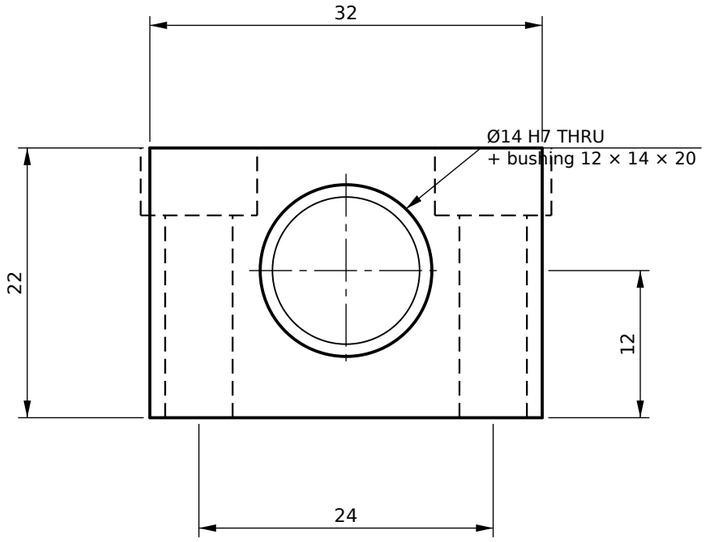


SECTION B–B (X = +30)

NOTES

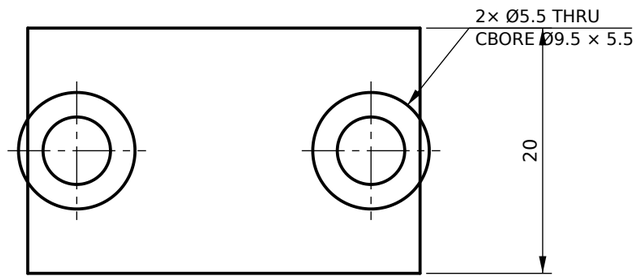
1. Rev B: width 80 (was 70) to carry the return rods at X = ±30 with full thread walls.
2. B1/B2 Ø32 H7 at 210.00 ±0.01 for flanged self-lubricating bronze bushings ID 25 / OD 32 / L 33; same program as MP-101/MP-108.
3. Top: clevis pocket 20.02 × 32.02 × 2.0 (R2) with V1/V2 M5 × 10; R1/R2 M10 × 15 (return rods); U1/U2 M6 × 10 (urethane bumpers).
4. Bottom: H Ø24 H7 × 4.2 (die-holder spigot); S1–S4 M5 × 10 on PCD 40 at 45°; Rev B: PL M6 × 12 for the station pilot MP-120.
5. Rev C.1: L1/L2 link-clearance pockets on the top, X ±10 to ±19, 15 long, 3 deep, R2, open into the clevis pocket. The R12 link ends dip 2 mm below the ram top.
6. Top and bottom faces parallel 0.02. Plan drawn with the front of the press to the right; hidden features are on the bottom.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 11 / 33
TITLE: **RAM**			
DRAWING NO.: **MP-109**	QTY: 1	SCALE: 1:1	UNITS: mm
MATERIAL: C45 flat 90 × 35, faced to 80 × 30			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:



FRONT VIEW (pin axis = X)

TOP VIEW

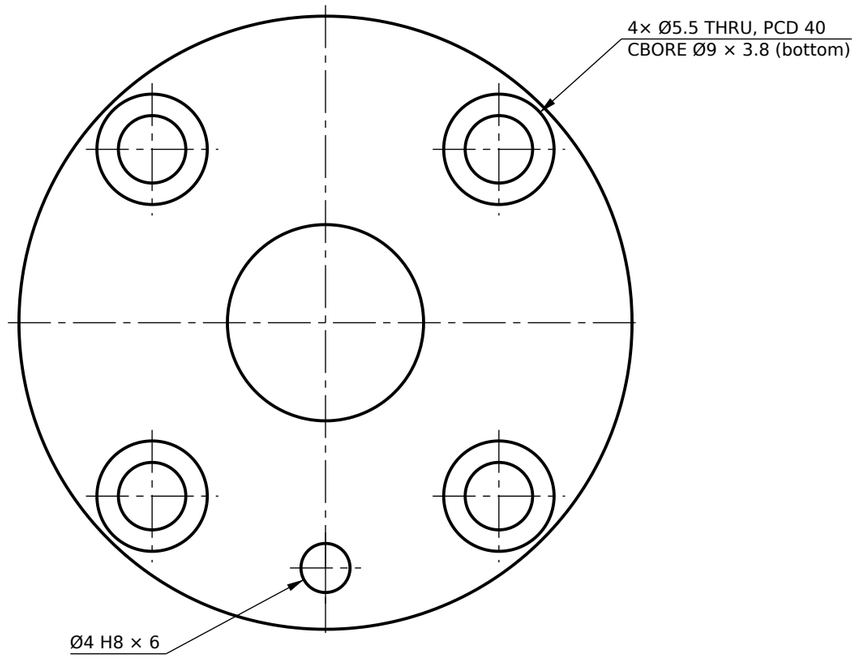


NOTES

1. Sits 2 mm deep in the ram pocket; 2 × M5 × 25 ISO 4762 + threadlocker.
2. Rev B: bore Ø14 H7 with a PTFE-composite plain bushing (DU / PAP type) 12 × 14 × 20 pressed in; pin B Ø12 runs dry.
3. Bore axis parallel to the bottom face within 0.02 and square to the long sides within 0.02.
4. Vertical edges chamfer 2.5 × 45° to clear the R2 pocket corners.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 12 / 33
TITLE: **CLEVIS**			
DRAWING NO.: **MP-110**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: C45 flat 25 × 35			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

SECTION THROUGH X AXIS

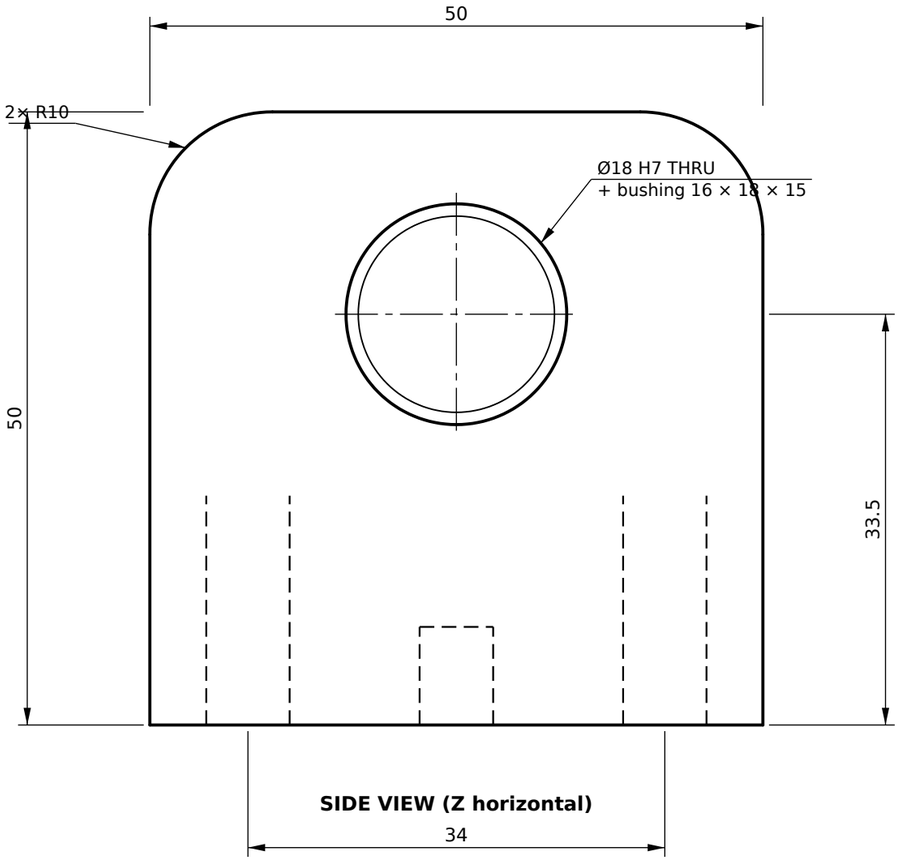


BOTTOM VIEW (front at bottom)

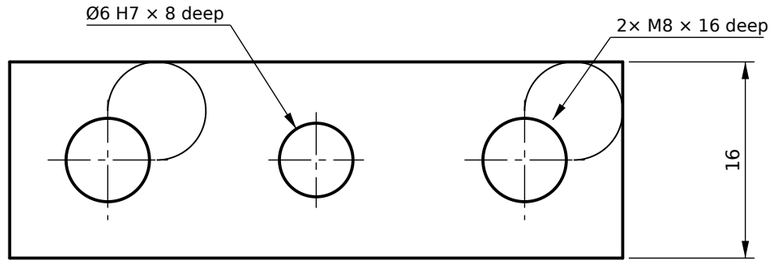
NOTES

1. Spigot Ø24 h6 locates in the ram (Ø24 H7). 4 × M5 × 16 DIN 7984 low-head screws from below + threadlocker.
2. Bore Ø16 H7 takes the upper-die shank MP-202. Bottom face (die seat) square to the bore within 0.01, flat 0.01.
3. Lock screw: M5 × 10 ISO 4027 cone-point set screw with nylon patch; seats in the shank V-groove. Check at every die change.
4. Orientation hole Ø4 H8 × 6 deep in the bottom face at 20 mm radius, on the Z axis toward the FRONT.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 13 / 33
TITLE: **DIE HOLDER**			
DRAWING NO.: **MP-111**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: 42CrMo4 Ø55 bar, QT 28–32 HRC			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN: / CHECKED:



SIDE VIEW (Z horizontal)



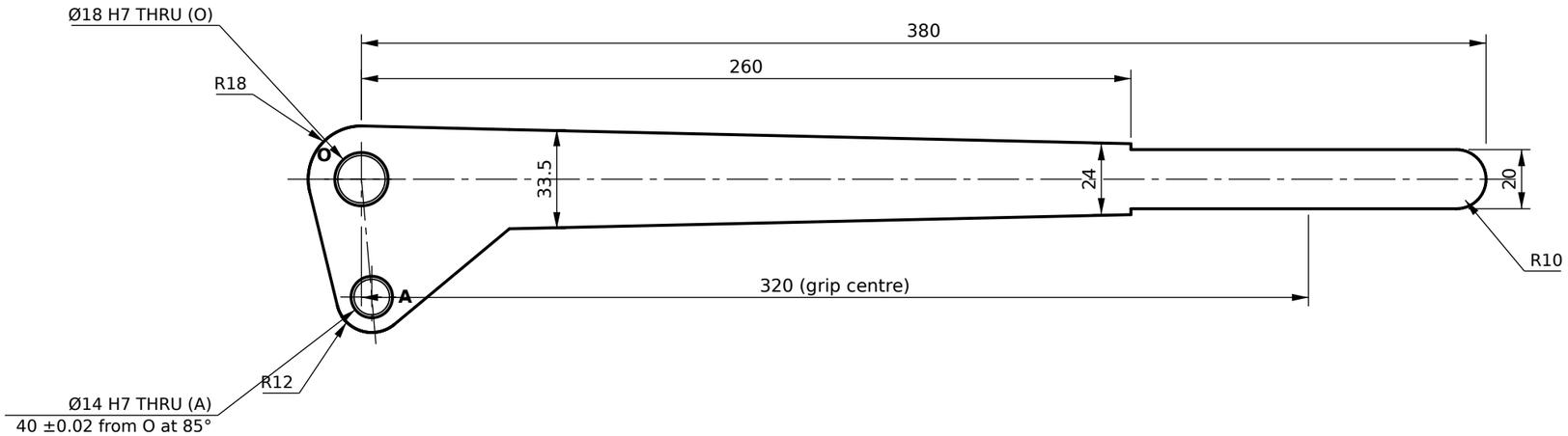
BOTTOM VIEW

NOTES

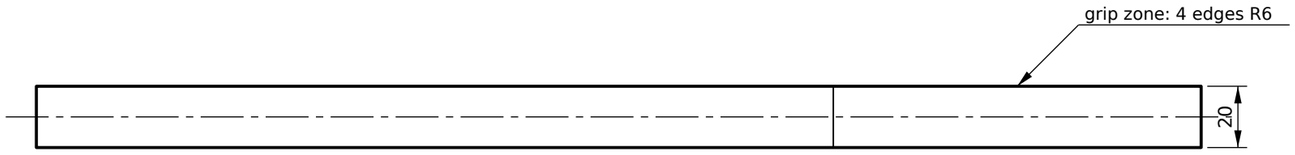
1. Rev B: pivot bore Ø18 H7 for a PTFE-composite plain bushing (DU / PAP type) 16 × 18 × 15 (pin O is now Ø16). Line-ream both cheeks together after fitting to the crown.
2. Fixed to the crown with 2 × M8 × 30 ISO 4762 + Nord-Lock washers from below + 1 × Ø6 × 16 dowel. Carries the full toggle reaction (16 kN design).
3. Bottom face flat 0.02, square to the sides.
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 14 / 33
TITLE: **PIVOT CHEEK**			
DRAWING NO.: **MP-112**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: C45 flat 20 × 55			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PROFILE (handle axis horizontal)



EDGE VIEW (thickness)

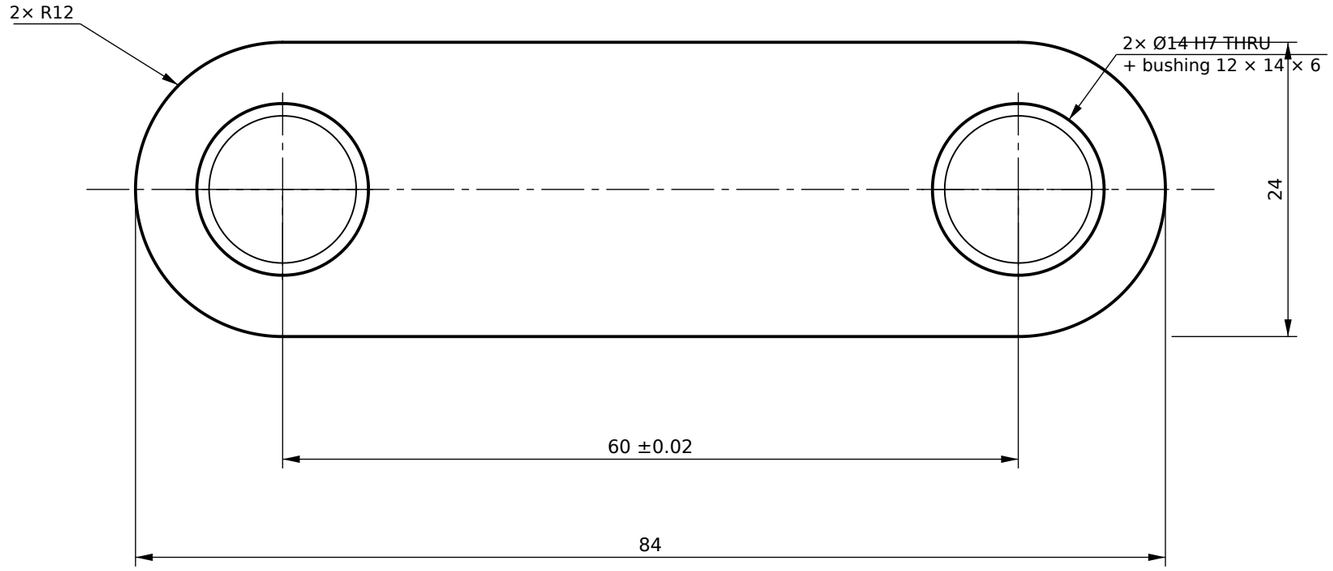


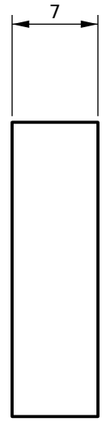
NOTES

1. Rev B: replaces the Rev A hub + screwed bar (MP-115 deleted). The Rev A M12 thread reached ≈ 465 MPa at a 200 N push; this section sees ≈ 14 MPa (36 MPa at 500 N).
2. Profile from DXF (laser/waterjet + CNC, or wire EDM). Faces parallel 0.05.
3. O: Ø18 H7 for a PTFE-composite plain bushing (DU / PAP type) 16 × 18 × 20. A: Ø14 H7 for a 12 × 14 × 20. Crank O→A = 40.00 ±0.02 at 85° ±0.1° to the handle axis.
4. Grip zone (u 260–380): 20 × 20 with all four long edges R6; red TPR grip Ø32 / Ø24 × 120 pushed on with rubber adhesive. Grip centre 320 from O.
5. Handle stop: the lever underside rests on the adjustable stop screw on the front column at 3° above horizontal.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 15 / 33
TITLE: **HANDLE LEVER (ONE PIECE)**			
DRAWING NO.: **MP-113**	QTY: 1	SCALE: 1:2	UNITS: mm
MATERIAL: C45 plate 25 mm, faced to 20			
HEAT TREATMENT / FINISH: Black oxide; grip zone edges R6			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PROFILE



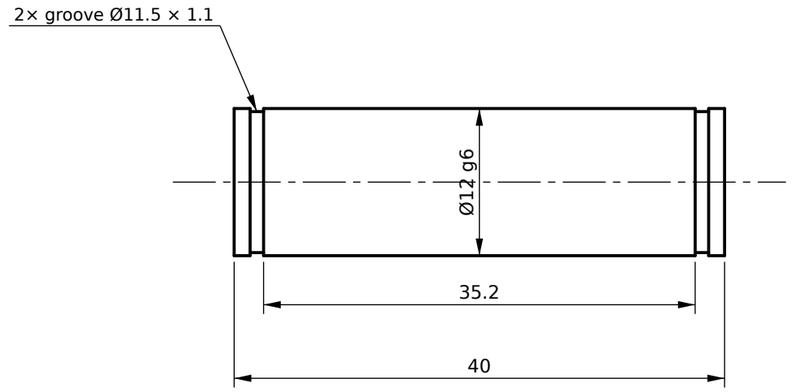


EDGE VIEW

NOTES

1. Hole centres 60.00 ±0.02; make both links as a matched pair (within 0.01).
2. Rev B: bores Ø14 H7 with PTFE-composite plain bushing (DU / PAP type) 12 × 14 × 6 pressed flush. Pins run dry.
3. Faces parallel 0.02. Stamp the pair with matching numbers.
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 16 / 33
TITLE: **TOGGLE LINK**			
DRAWING NO.: **MP-114**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: C45 plate 8 mm, faced to 7			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:

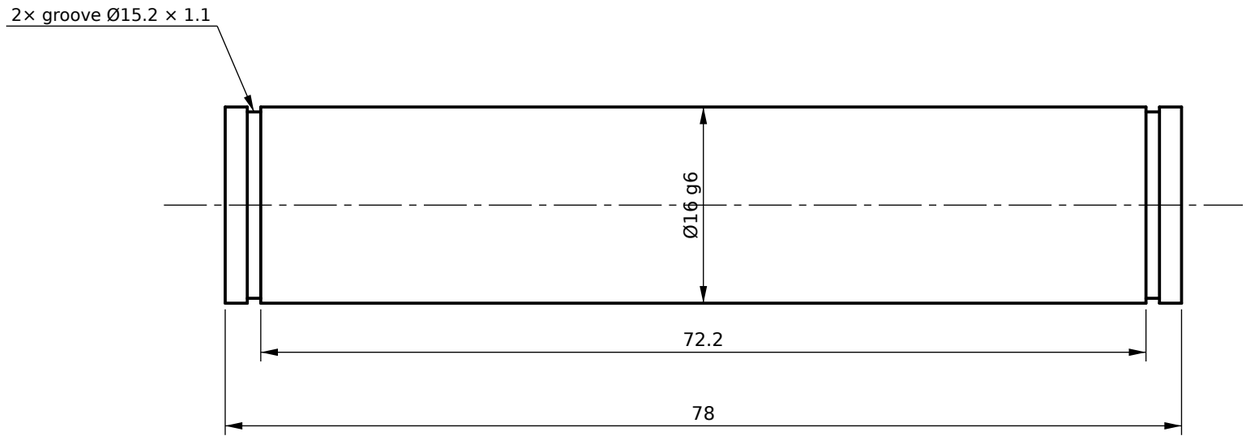


SIDE VIEW

NOTES

1. Retaining-ring grooves for DIN 471 – 12: groove Ø11.5 h12, width 1.1 H13, both ends.
2. Ends 0.5 × 45° chamfer. Groove inner edge at 17.6 from the pin centre (bearing faces at ±17.5).
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 17 / 33
TITLE: **PIN A / PIN B (TOGGLE)**			
DRAWING NO.: **MP-116**	QTY: 2	SCALE: 2:1	UNITS: mm
MATERIAL: 100Cr6 (GCr15) bar, or hardened ground dowel stock Ø12 g6			
HEAT TREATMENT / FINISH: Harden 58–62 HRC, grind Ø12 g6, Ra 0.4. Runs in PTFE-composite bushings: no grease			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

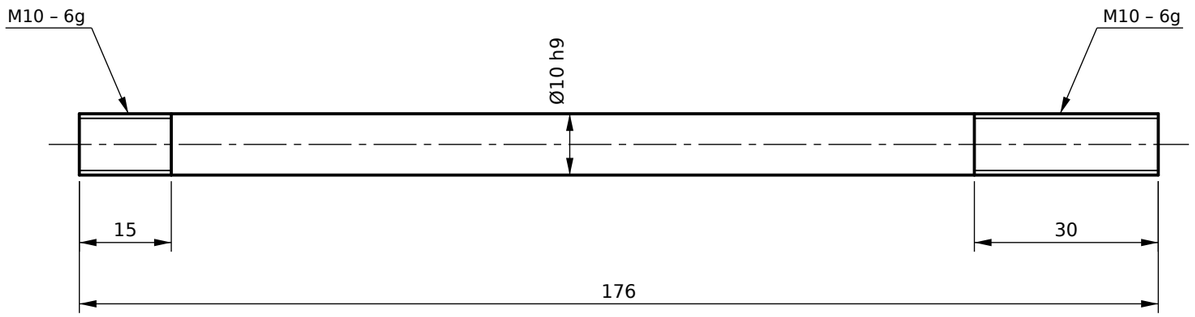


SIDE VIEW

NOTES

1. Retaining-ring grooves for DIN 471 – 16: groove Ø15.2 h12, width 1.1 H13, both ends.
2. Ends 0.5 × 45° chamfer. Groove inner edge at 36.1 from the pin centre (bearing faces at ±36).
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 18 / 33
TITLE: **PIN O (HANDLE PIVOT)**			
DRAWING NO.: **MP-117**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: 100Cr6 (GCr15) bar, or hardened ground dowel stock Ø16 g6			
HEAT TREATMENT / FINISH: Harden 58–62 HRC, grind Ø16 g6, Ra 0.4. Runs in PTFE-composite bushings: no grease			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

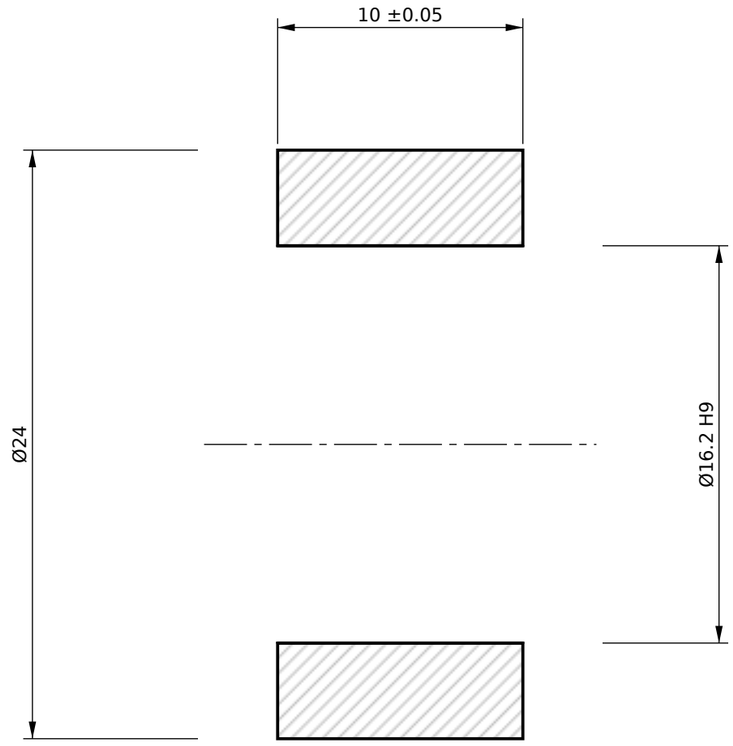


SIDE VIEW

NOTES

1. Bottom thread screws into the ram MP-109 (M10 × 15) with Loctite 243.
2. Return spring: music wire Ø3.5 (EN 10270-1 DH), OD 31.5, free length 128, ≈ 16.4 active coils, ends closed and ground, shot-peened.
3. Rate ≈ 4.25 N/mm. Installed 109.5 at handle-up (≈ 80 N); 70.6 at handle-down (≈ 245 N, τ ≈ 480 MPa). Solid length ≤ 64.5.
4. Top: Ø32 × 3 washer and M10 nylock nut. The two springs together must lift the handle from the fully-down position.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 19 / 33
TITLE: **RETURN ROD**			
DRAWING NO.: **MP-118**	QTY: 2	SCALE: 1:1	UNITS: mm
MATERIAL: C45 Ø12 bright bar			
HEAT TREATMENT / FINISH: Zinc plate 8 µm			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

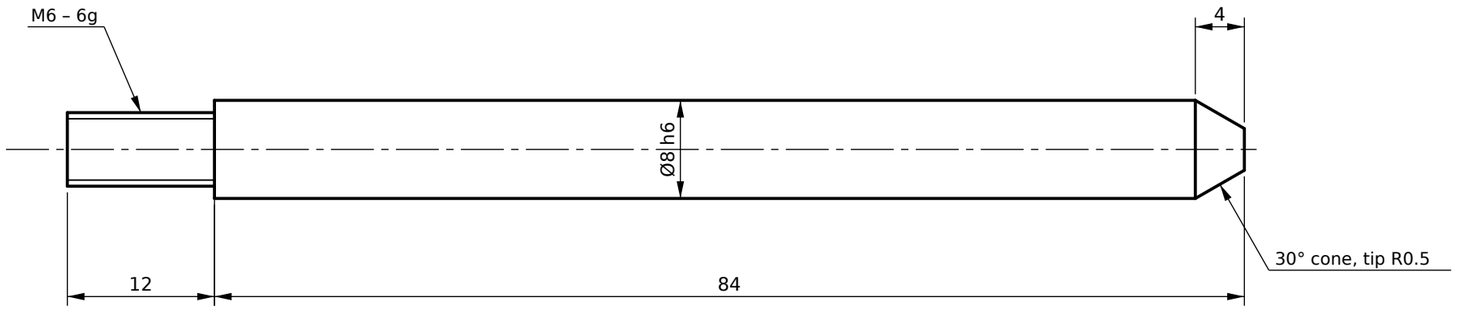


SECTION

NOTES

1. Rev B: bore Ø16.2 for the Ø16 pin O. Fits between the handle lever and each pivot cheek.
2. Faces parallel within 0.02.
3. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
4. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 20 / 33
TITLE: **PIVOT SPACER**			
DRAWING NO.: **MP-119**	QTY: 2	SCALE: 4:1	UNITS: mm
MATERIAL: CuSn8 bronze tube Ø26 × Ø14, or bar			
HEAT TREATMENT / FINISH: None			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED



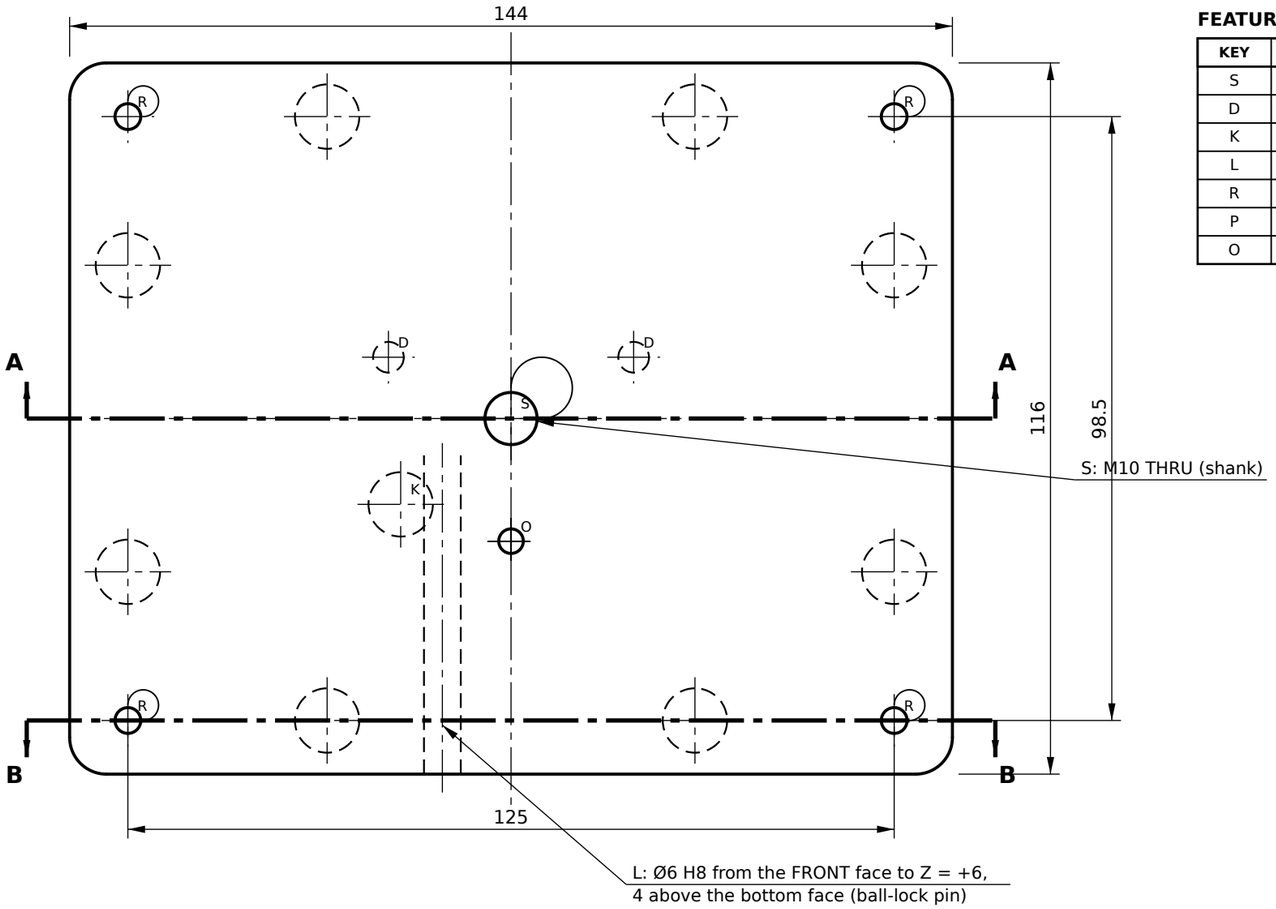
SIDE VIEW

NOTES

1. NEW in Rev B. Screwed into the ram underside (MP-109, hole PL) with Loctite 243; tip 84 below the ram.
2. Enters the die table hole PA or PB 6 mm before the dies touch. If the table is more than about 2 mm off a station, the tip lands on the table and stops the stroke with the dies still 6 mm apart.
3. Within ±2 mm, the 30° tip pulls the table onto the exact station (±0.02).
4. Tip: 30° half-angle cone, 4 long, tip radius R0.5, polished.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **C.1**	SHEET: 21 / 33	
TITLE: **STATION PILOT**			
DRAWING NO.: **MP-120**	QTY: 1	SCALE: 2:1	UNITS: mm
MATERIAL: 100Cr6 (GCr15) Ø10 bar			
HEAT TREATMENT / FINISH: Harden 58–62 HRC, grind Ø8 h6, Ra 0.4			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom



FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
S	centre	M10 THRU (shank MP-202, Loctite 243)
D	(±20, −10)	Ø5 H7 × 6 from the BOTTOM (plug dowels)
K	(−18, +14)	Ø10.5 × 8 from the BOTTOM (plug stud MP-212)
L	X −11.2, Z +58 → +6	Ø6 H8 along Z, 4 above the bottom face (ball-lock pin Ø6, grip ≥ 55)
R	(±62.5, ±49.25)	M5 THRU (carrier shoulder screws ISO 7379 Ø6 × 10)
P	(±30, ±49.25), (±62.5, ±25)	Ø10.5 × 8 from the BOTTOM (8 upper springs)
O	(0, +20)	Ø4 H7 × 6 from the TOP (orientation pin Ø4 m6 × 10)



SECTION A–A

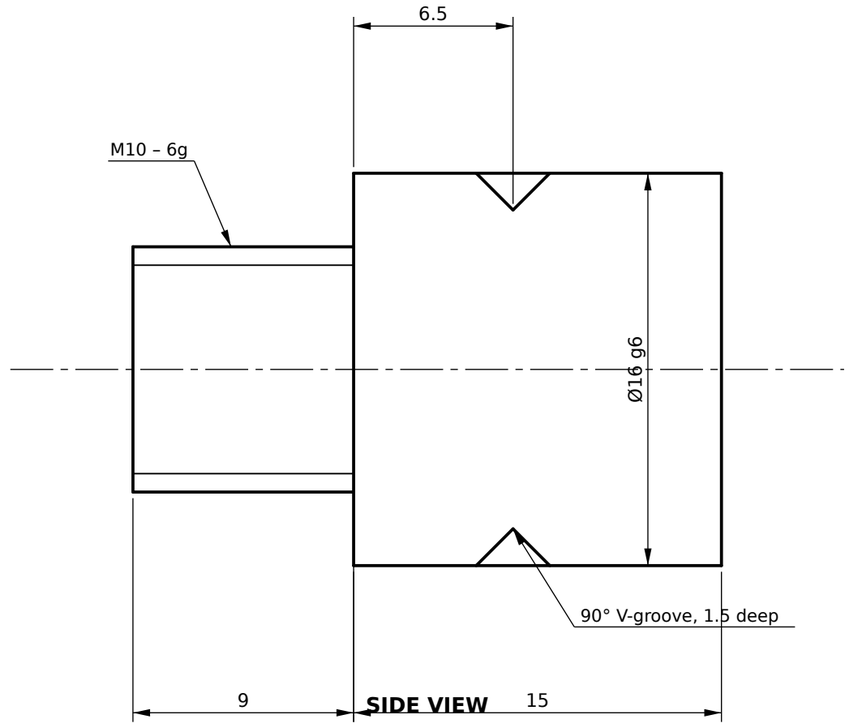


SECTION B–B (Z = ±49.25)

NOTES

1. Universal die part: one per press, the same for every magnet size. X = along the die table, +Z = front.
2. Laser-mark the drawing number on a side face.
3. Rev C: one body per press (144 × 116). The plug now hangs on dowels D and the stud K, held by a Ø6 ball-lock pin in hole L: no screws to change a size.
4. Top face (seat against the holder through the shim pack) and bottom face parallel 0.01, flat 0.01. The shim pack sets the crimp depth once for all kits.
5. Upper springs: 8 × ISO 10243 Ø10 × 32 blue in 26 mm cavities: 6 mm preload (≈ 0.68 kN total), ≈ 1.3 kN at full retraction (36%). Replace every 250,000 strokes.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press			REVISION: **C.1**	SHEET: 22 / 33
TITLE: **UPPER DIE BODY (UNIVERSAL)**				
DRAWING NO.: **MP-201**	QTY: 1 per press	SCALE: 1:1	UNITS: mm	
MATERIAL: 1.2312 (P20+S) prehardened 28–32 HRC				
HEAT TREATMENT / FINISH: Black oxide				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:



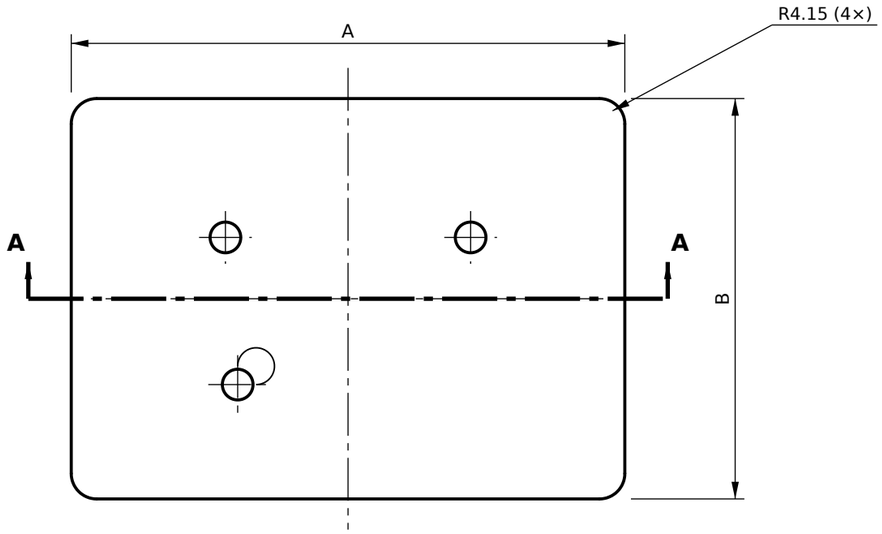
SIDE VIEW

NOTES

1. Screw into the die body MP-201 (M10 THRU) with Loctite 243 until the Ø16 face seats; face square to axis 0.01.
2. V-groove takes the cone-point lock screw in the die holder MP-111 (groove centre 6.5 from the seating face).
3. Top end chamfer 1 × 45° (lead-in).
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 23 / 33
TITLE: **UPPER DIE SHANK**			
DRAWING NO.: **MP-202**	QTY: 1 per press	SCALE: 4:1	UNITS: mm
MATERIAL: 42CrMo4 Ø20 bar, QT 28–32 HRC			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:

PLAN (TOP) — front at bottom



DIMENSION TABLE (mm)

SIZE	X	Z	A	B
5050	50	50	50.3	50.3
6363	63.5	63.5	63.8	63.8
5380	80	53	80.3	53.3
6590	90	65	90.3	65.3

Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X + 0.30 (plug length, ±0.01)

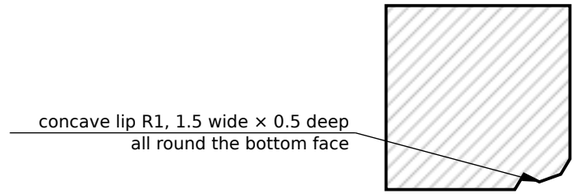
B = Z + 0.30 (plug width, ±0.01)

FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
S	(−18, +14)	M6 × 10 from the TOP (stud MP-212, Loctite 243)
D	(±20, −10)	Ø5 H7 × 8 from the TOP (dowels Ø5 m6 × 14)



SECTION A–A



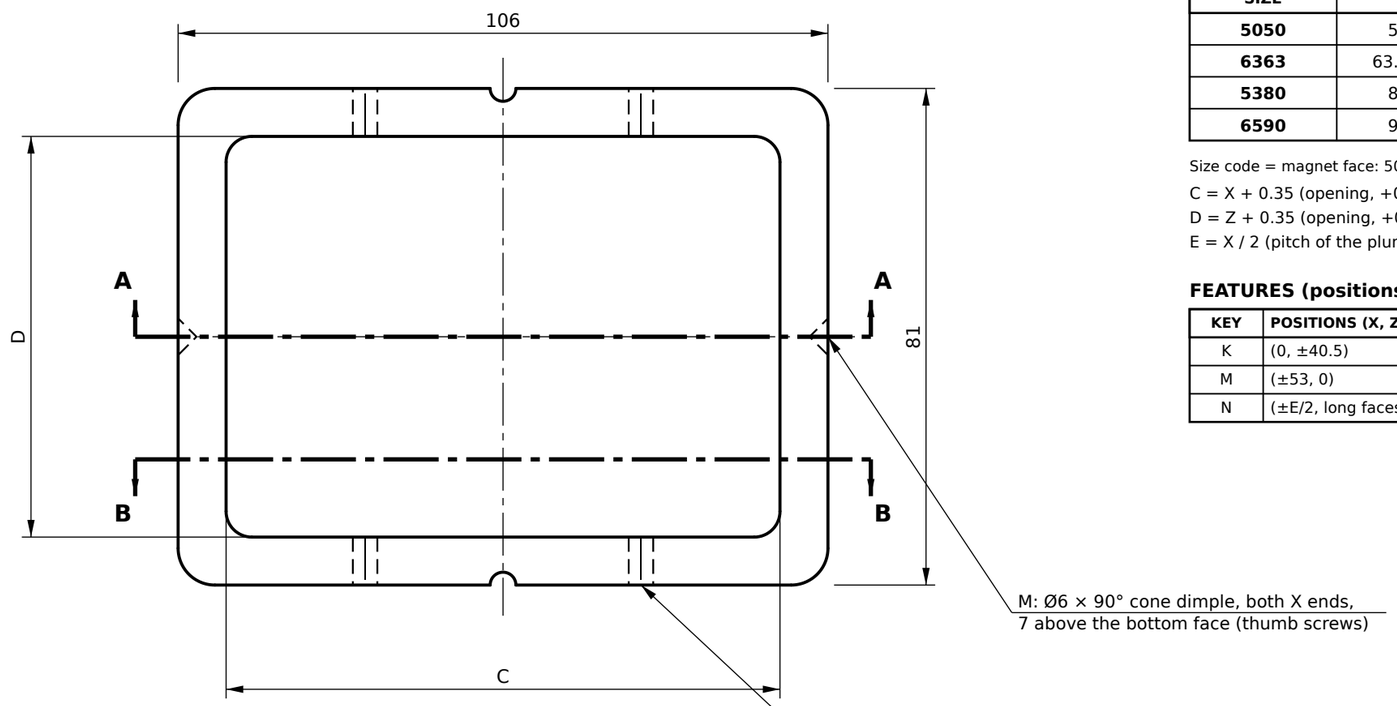
DETAIL Y (5:1) working edge

NOTES

1. Size kit part: one per magnet size, tabulated below. Shown to scale for size 6590 (65 × 90). X = long side, along the die table.
2. Laser-mark on a side face: drawing no., size code and U / A / B where applicable (e.g. "MP-207 6590 A").
3. Rev C: no screws. Two Ø5 m6 × 14 dowels pressed in (5.5 protruding) locate it in the body; stud MP-212 in the M6 hole takes the ball-lock pin.
4. Profile A × B, R4.15: wire-EDM after hardening, ±0.01. Sliding fit in the upper ring (ring = plug + 0.05).
5. Bottom (working) face flat 0.01, Ra 0.2, lip per Detail Y. Height 14.5 ±0.01 sets the 5.5 recess in the ring.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **C.1**	SHEET: 24 / 33	
TITLE: **PLUG (SIZE KIT)**			
DRAWING NO.: **MP-203**	QTY: 1 per size kit (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype kit; PM cold-work steel for high volume			
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom



DIMENSION TABLE (mm)

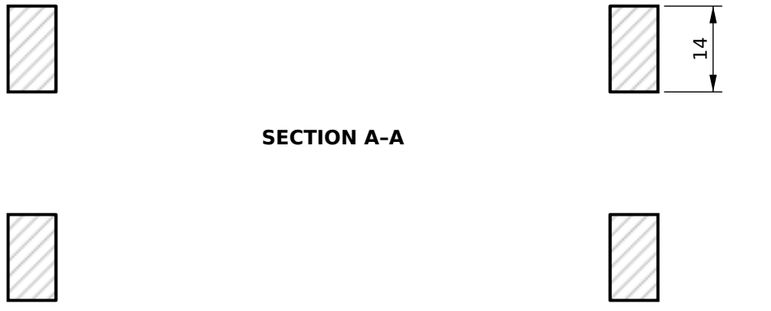
SIZE	X	Z	C	D	E
5050	50	50	50.35	50.35	25
6363	63.5	63.5	63.85	63.85	31.75
5380	80	53	80.35	53.35	40
6590	90	65	90.35	65.35	45

Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

C = X + 0.35 (opening, +0.01/0)
D = Z + 0.35 (opening, +0.01/0)
E = X / 2 (pitch of the plungers in X)

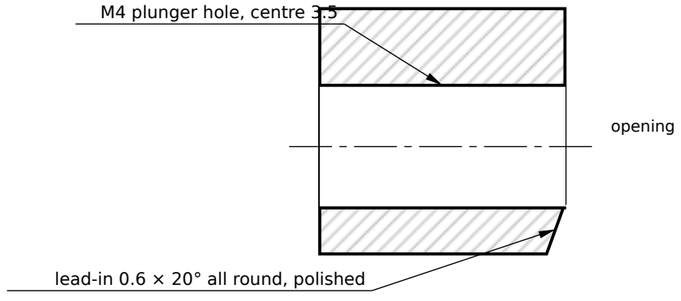
FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
K	(0, ±40.5)	Key slot, half-round Ø4.2, full height, on both long faces
M	(±53, 0)	Ø6 × 90° cone dimple, 1.5 deep, 7 above the bottom face
N	(±E/2, long faces)	M4 radial THRU, 3.5 above the bottom face (spring plungers)



SECTION A–A

SECTION B–B (Z = ±20)



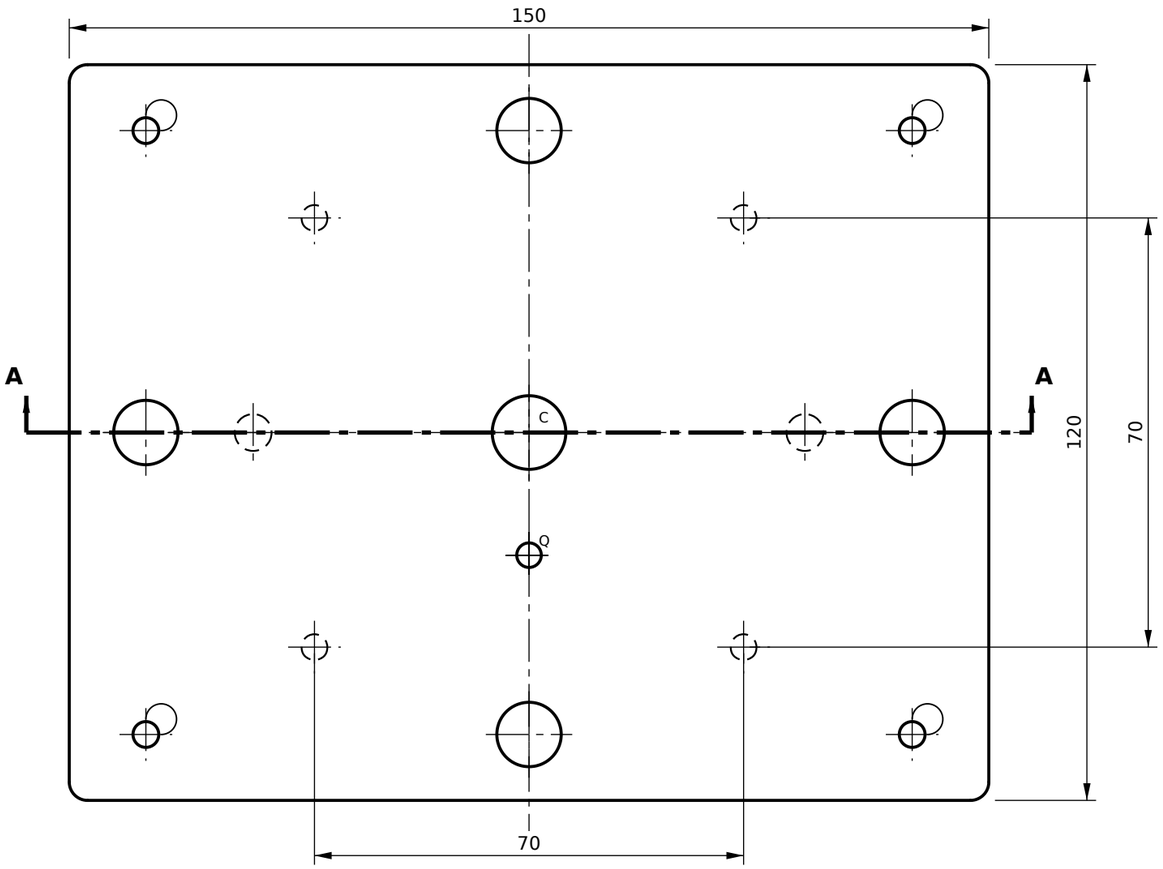
DETAIL P (5:1) bottom inner edge

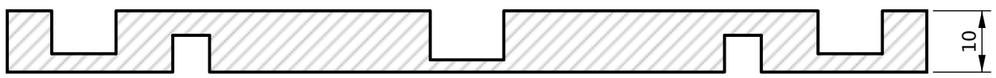
NOTES

1. Size kit part: one per magnet size, tabulated below. Shown to scale for size 6590 (65 × 90). X = long side, along the die table.
2. Laser-mark on a side face: drawing no., size code and U / A / B where applicable (e.g. "MP-207 6590 A").
3. Outside 106 × 81 (−0.01/−0.02) for every size: fits the upper carrier MP-210 pocket. Key slots K (Ø4.2) take the carrier pins.
4. Opening C × D, R4.2, THRU: wire-EDM after hardening, lap Ra 0.2. Bottom inner lead-in 0.6 × 20° (Detail P).
5. Held by 2 cone-point thumb screws in dimples M. Four nylon-tip spring plungers N protrude 0.2–0.3 into the opening; body length ≤ wall − 0.5.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **C.1**	SHEET: 25 / 33	
TITLE: **UPPER RING (SIZE KIT)**			
DRAWING NO.: **MP-204**	QTY: 1 per size kit (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype kit; PM cold-work steel for high volume			
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom





SECTION A–A

FEATURES (positions from the part centre, +Z = front)

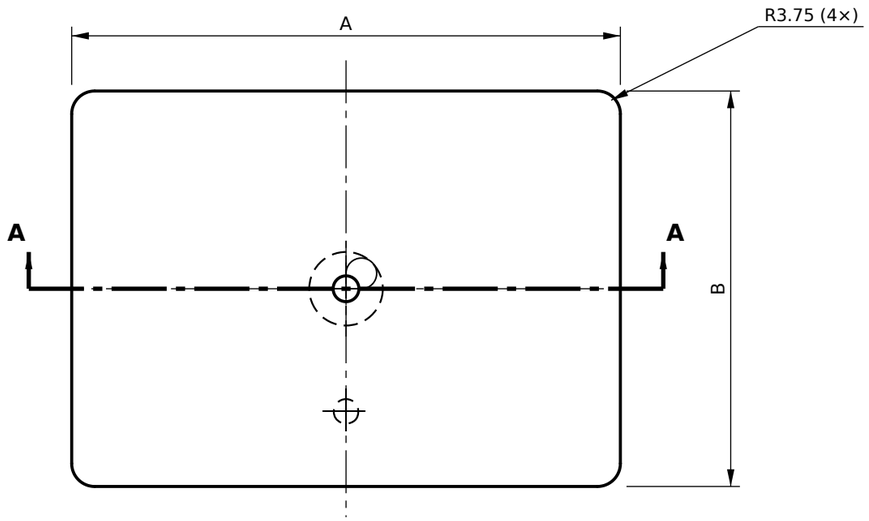
KEY	POSITIONS (X, Z)	FEATURE
C	centre	Ø12 H7 × 8 from the TOP (pedestal / anvil spigot)
Q	(0, +20)	Ø4 H8 × 6 from the TOP (anti-rotation pin)
P	(±62.5, 0), (0, ±49.25)	Ø10.5 × 7 from the TOP (4 lower springs)
R	(±62.5, ±49.25)	M5 THRU (carrier shoulder screws ISO 7379 Ø6 × 12)
F	(±35, ±35)	M5 × 8 from the BOTTOM (fixing to the die table, M5 × 16 ISO 10642 from below)
G	(±45, 0)	Ø6 H7 × 6 from the BOTTOM (dowels Ø6 m6 × 12 into the table)

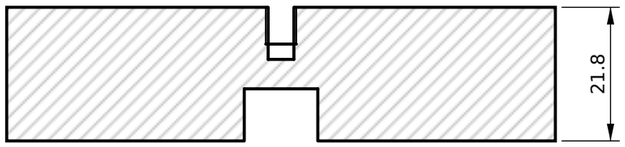
NOTES

1. Universal die part: one per press, the same for every magnet size. X = along the die table, +Z = front.
2. Laser-mark the drawing number on a side face.
3. Rev C: replaces the Rev B shoes. Both stations use this part; it stays fixed to the die table and is never removed for a size change.
4. Top and bottom faces parallel 0.01, flat 0.01.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press			REVISION: **C.1**	SHEET: 26 / 33
TITLE: **STATION BASE (UNIVERSAL)**				
DRAWING NO.: **MP-205**	QTY: 2 per press	SCALE: 1:1	UNITS: mm	
MATERIAL: C45 plate 12 mm, faced to 10				
HEAT TREATMENT / FINISH: Black oxide				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN	CHECKED

PLAN (TOP) — front at bottom





SECTION A–A

DIMENSION TABLE (mm)

SIZE	X	Z	A	B
5050	50	50	49.5	49.5
6363	63.5	63.5	63	63
5380	80	53	79.5	52.5
6590	90	65	89.5	64.5

Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

A = X − 0.5 (±0.01)

B = Z − 0.5 (±0.01)

FEATURES (positions from the part centre, +Z = front)

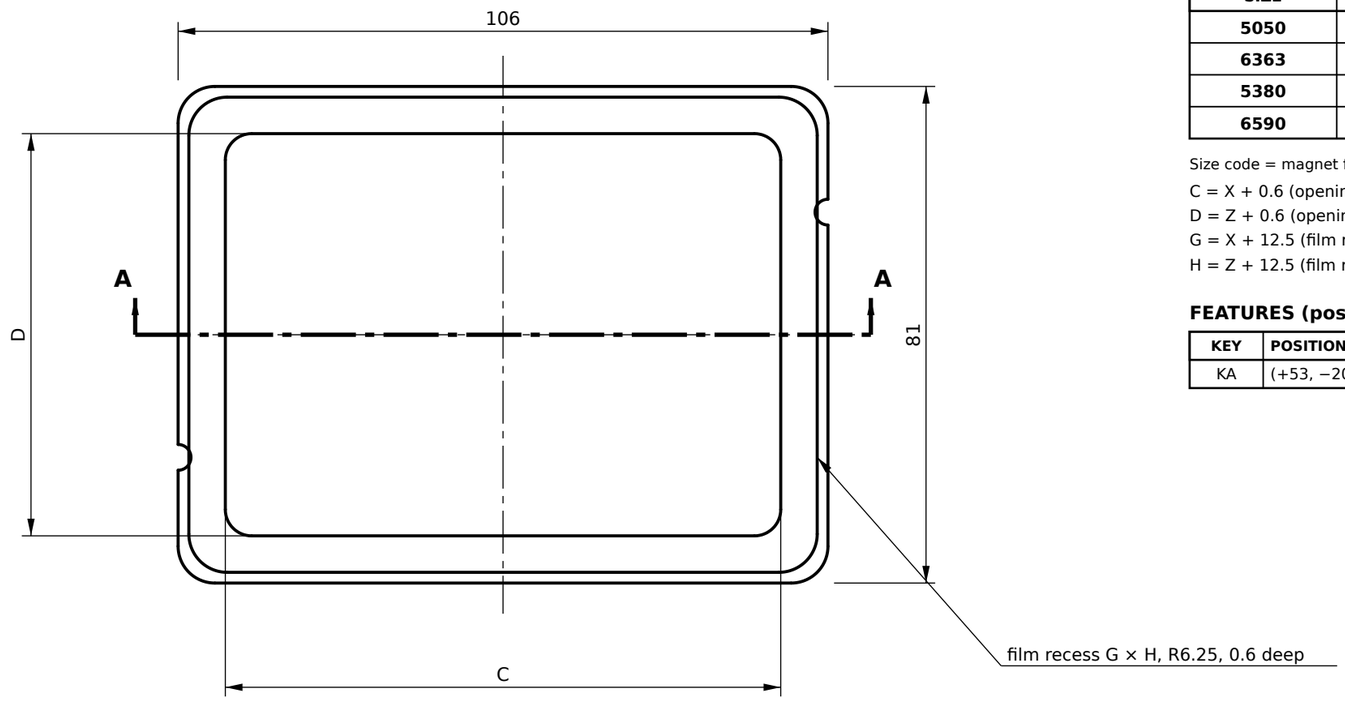
KEY	POSITIONS (X, Z)	FEATURE
H	centre	M5 × 6 from the TOP (puller)
C	centre	Ø12 H7 × 8.5 from the BOTTOM (spigot dowel)
Q	(0, +20)	Ø4 H7 × 6 from the BOTTOM (anti-rotation pin)

NOTES

1. Size kit part: one per magnet size, tabulated below. Shown to scale for size 6590 (65 × 90). X = long side, along the die table.
2. Laser-mark on a side face: drawing no., size code and U / A / B where applicable (e.g. "MP-207 6590 A").
3. The tinplate shell sits face-up over it. Top face flat 0.01, Ra 0.2; top edge R0.3. Height 21.8 ±0.02.
4. Rev C: drop-in. Press a Ø12 m6 × 15 dowel into C (7 protruding) and a Ø4 m6 × 10 pin into Q (4 protruding). Lift out with the M5 T-handle puller in hole H.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 27 / 33	
TITLE: **PICKUP PEDESTAL (SIZE KIT)**				
DRAWING NO.: **MP-206**	QTY: 1 per size kit (4)	SCALE: 1:1 (size 6590)	UNITS: mm	
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype kit; PM cold-work steel for high volume				
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:

PLAN (TOP) — front at bottom



DIMENSION TABLE (mm)

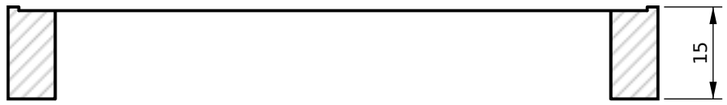
SIZE	X	Z	C	D	G	H
5050	50	50	50.6	50.6	62.5	62.5
6363	63.5	63.5	64.1	64.1	76	76
5380	80	53	80.6	53.6	92.5	65.5
6590	90	65	90.6	65.6	102.5	77.5

Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

C = X + 0.6 (opening)
D = Z + 0.6 (opening)
G = X + 12.5 (film recess, 0.6 deep)
H = Z + 12.5 (film recess)

FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
KA	(+53, −20), (−53, +20)	Key slot, half-round Ø4.2, full height, on the short faces



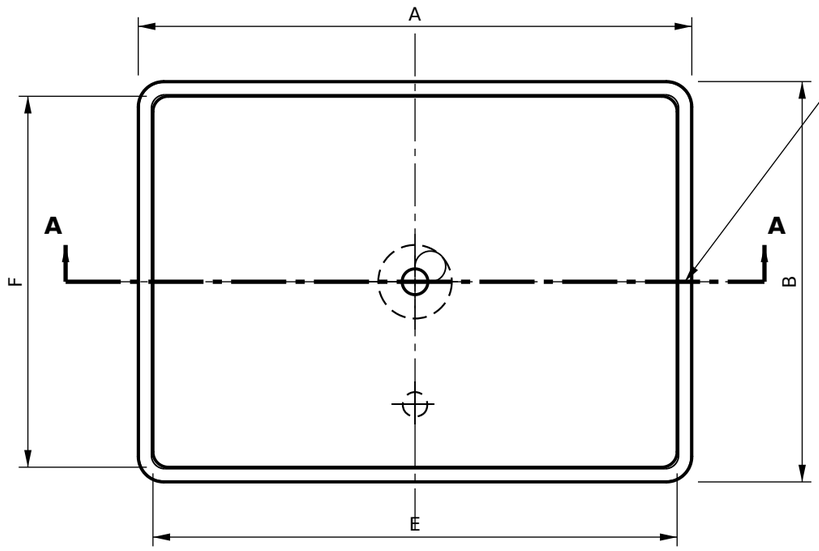
SECTION A–A

NOTES

1. Size kit part: one per magnet size, tabulated below. Shown to scale for size 6590 (65 × 90). X = long side, along the die table.
2. Laser-mark on a side face: drawing no., size code and U / A / B where applicable (e.g. "MP-207 6590 A").
3. Outside 106 × 81 (−0.01/−0.02) for every size: drops into the lower carrier A pocket. Key slots KA fit only carrier A.
4. Opening C × D, R4.3, THRU, wire-EDM; top inner edge R0.5. Film recess G × H, R6.25, 0.6 deep, Ra 0.8: it centres the photo and film.
5. Lift out by hand through the opening once the pedestal is out. Faces parallel 0.01.
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **C.1**	SHEET: 28 / 33	
TITLE: **PICKUP RING A (SIZE KIT)**			
DRAWING NO.: **MP-207**	QTY: 1 per size kit (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype kit; PM cold-work steel for high volume			
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom



DIMENSION TABLE (mm)

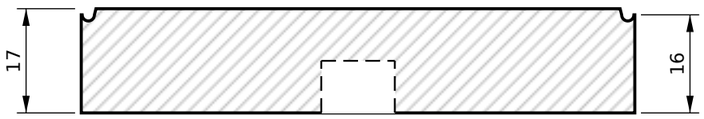
SIZE	X	Z	A	B	E	F
5050	50	50	50.3	50.3	45.5	45.5
6363	63.5	63.5	63.8	63.8	59	59
5380	80	53	80.3	53.3	75.5	48.5
6590	90	65	90.3	65.3	85.5	60.5

Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

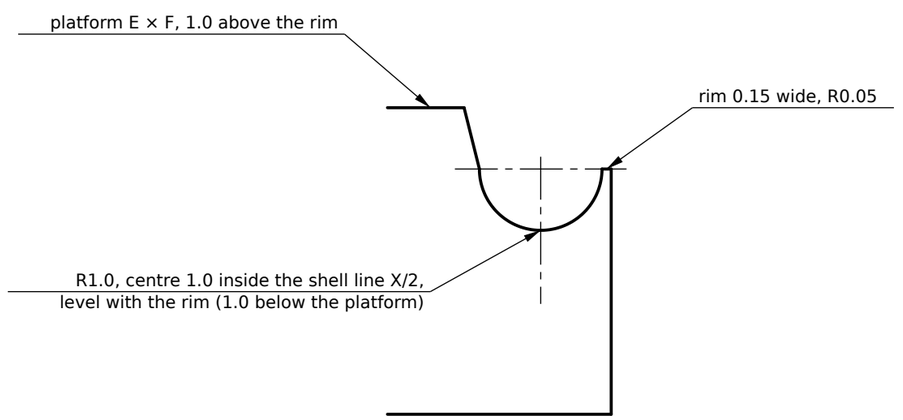
A = X + 0.3 (block, ±0.01)
B = Z + 0.3 (block, ±0.01)
E = X − 4.5 (platform)
F = Z − 4.5 (platform)

FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
H	centre	M5 × 6 from the TOP (puller)
C	centre	Ø12 H7 × 8.5 from the BOTTOM (spigot dowel)
Q	(0, +20)	Ø4 H7 × 6 from the BOTTOM (anti-rotation pin)



SECTION A–A



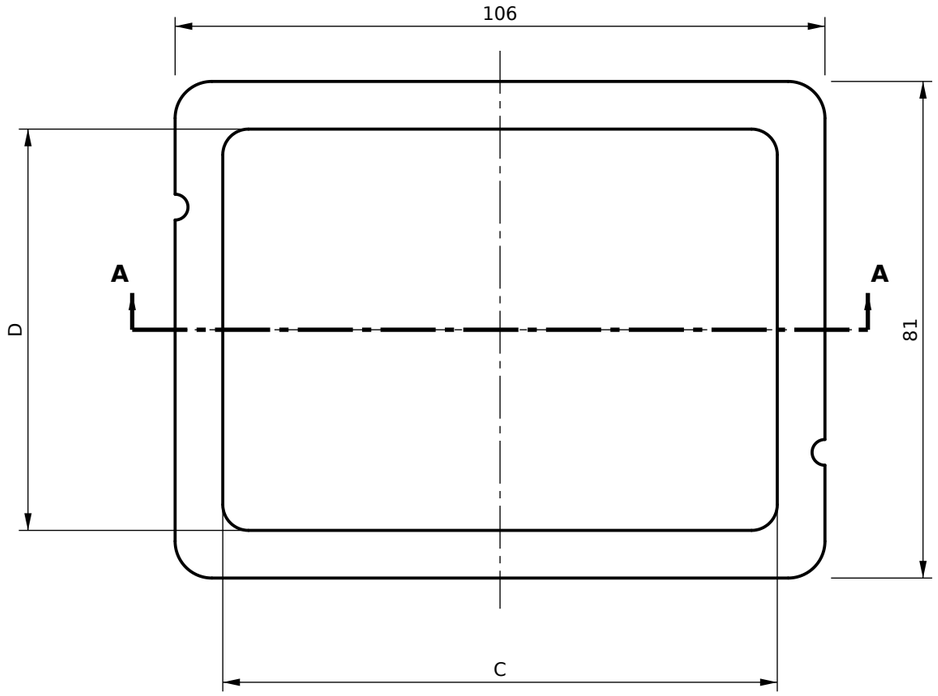
DETAIL R (10:1) roll channel — X/2 at the right edge of the rim

NOTES

1. Size kit part: one per magnet size, tabulated below. Shown to scale for size 6590 (65 × 90). X = long side, along the die table.
2. Laser-mark on a side face: drawing no., size code and U / A / B where applicable (e.g. "MP-207 6590 A").
3. Rev C: replaces the flat crimp pedestal. The roll channel (Detail R) turns the shell edge inward under the back plate; this is the crimp.
4. Block X + 0.3 slides in the crimp ring (X + 0.45) and guides it. Platform (X − 4.5) carries the back plate; the back overhangs it by 1.55 each side.
5. Channel profile is a start value: prove it on the 7075 prototype kit, then cut tool steel. Polish the channel Ra 0.2.
6. Drop-in: Ø12 m6 × 15 dowel in C (7 protruding), Ø4 m6 × 10 pin in Q. Lift out with the M5 puller in hole H.
7. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
8. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 29 / 33	
TITLE: **CRIMP ANVIL (SIZE KIT)**				
DRAWING NO.: **MP-208**	QTY: 1 per size kit (4)	SCALE: 1:1 / 10:1	UNITS: mm	
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype kit; PM cold-work steel for high volume				
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN	CHECKED

PLAN (TOP) — front at bottom



DIMENSION TABLE (mm)

SIZE	X	Z	C	D
5050	50	50	50.45	50.45
6363	63.5	63.5	63.95	63.95
5380	80	53	80.45	53.45
6590	90	65	90.45	65.45

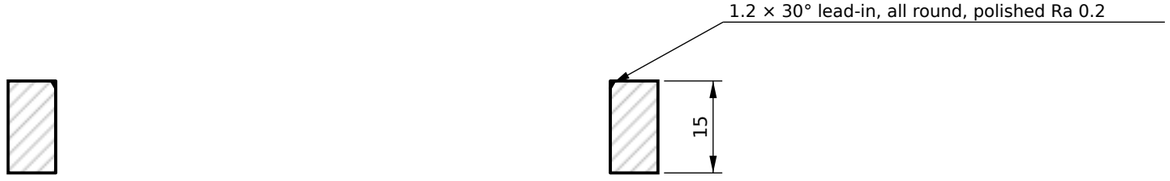
Size code = magnet face: 5050 = 50 × 50, 6363 = 63.5 × 63.5, 5380 = 53 × 80, 6590 = 65 × 90.

C = X + 0.45 (opening, +0.01/0)

D = Z + 0.45 (opening, +0.01/0)

FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
KB	(+53, +20), (−53, −20)	Key slot, half-round Ø4.2, full height, on the short faces



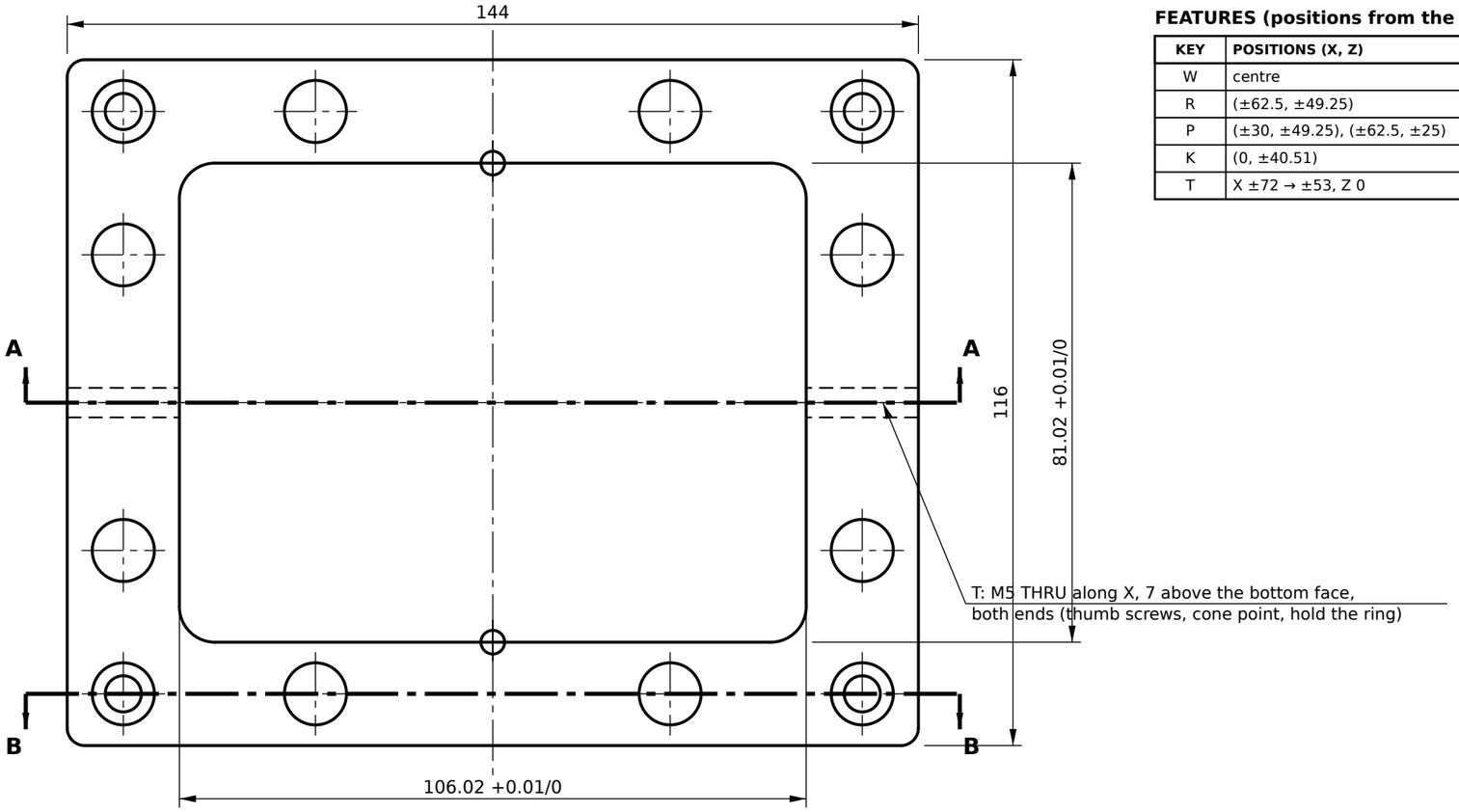
SECTION A–A

NOTES

1. Size kit part: one per magnet size, tabulated below. Shown to scale for size 6590 (65 × 90). X = long side, along the die table.
2. Laser-mark on a side face: drawing no., size code and U / A / B where applicable (e.g. "MP-207 6590 A").
3. Outside 106 × 81 (−0.01/−0.02) for every size: drops into the lower carrier B pocket. Key slots KB fit only carrier B.
4. Opening C × D, R4.2, THRU, wire-EDM, lap Ra 0.2. Top lead-in 1.2 × 30° folds the film under the shell edge. It guides the shell down onto the anvil channel.
5. A drop of silicone release on the lead-in stops laser-printed film slipping. No EDM recast on the forming edges (2 skim cuts + re-temper).
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press	REVISION: **C.1**	SHEET: 30 / 33	
TITLE: **CRIMP RING B (SIZE KIT)**			
DRAWING NO.: **MP-209**	QTY: 1 per size kit (4)	SCALE: 1:1 (size 6590)	UNITS: mm
MATERIAL: A2 / 1.2363 tool steel; 7075-T6 for the prototype kit; PM cold-work steel for high volume			
HEAT TREATMENT / FINISH: Harden 58–60 HRC, temper ×2; wire-EDM + 2 skim cuts, then re-temper; forming edges Ra 0.2; thin dense chrome			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom



FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
W	centre	Pocket 106.02 × 81.02 (+0.01/0), R6, THRU: holds the upper ring MP-204
R	(±62.5, ±49.25)	Ø6.1 THRU, CBORE Ø10.5 × 10 from the BOTTOM
P	(±30, ±49.25), (±62.5, ±25)	Ø10.5 × 12 from the TOP (8 upper springs)
K	(0, ±40.51)	Ø4 H7 THRU on the pocket wall; press in Ø4 m6 × 14 flush (key pins, upper ring only)
T	X ±72 → ±53, Z 0	M5 THRU along X, 7 above the bottom face (2 cone-point thumb screws)



SECTION A–A



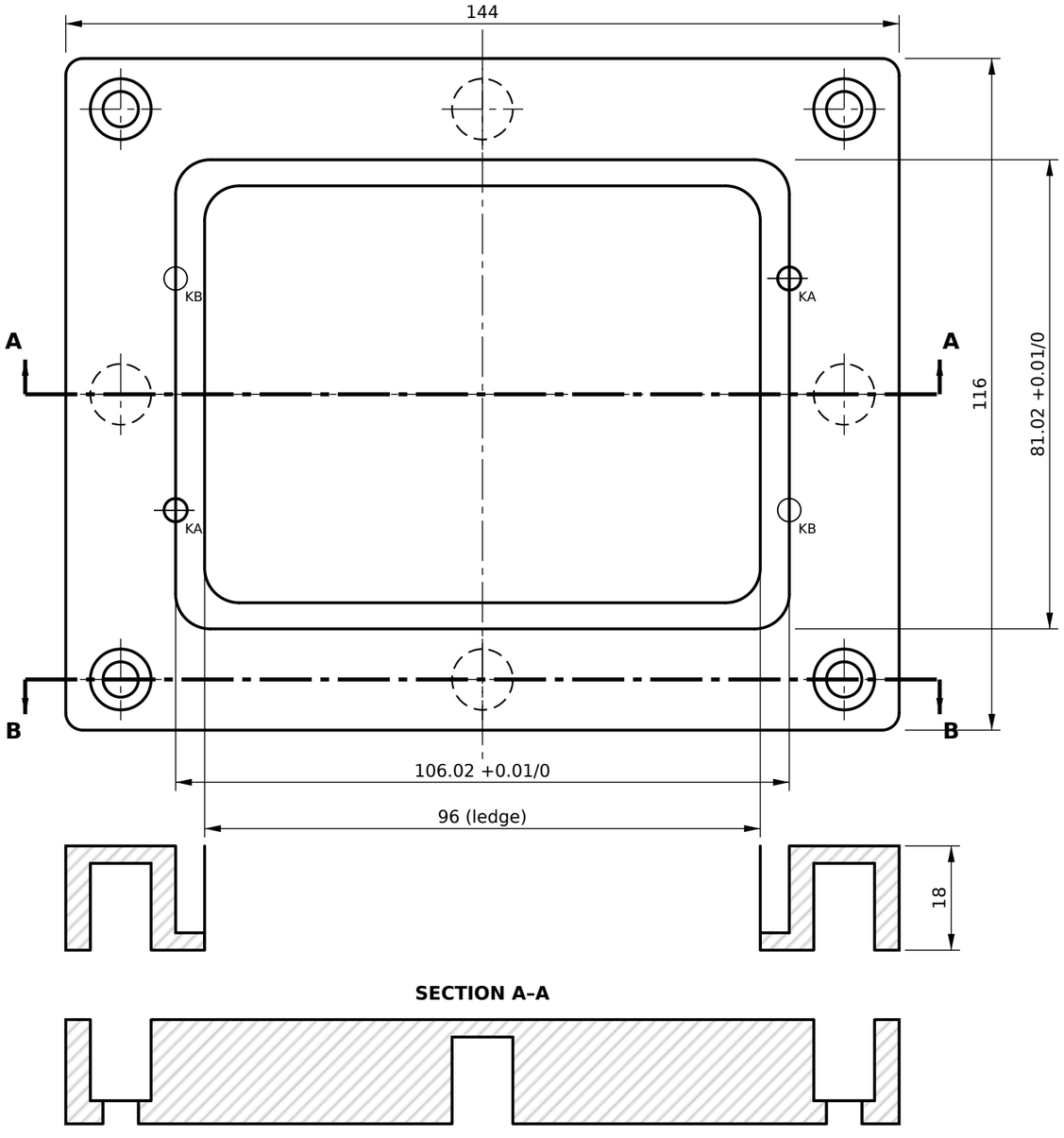
SECTION B–B (Z = ±49.25)

NOTES

1. Universal die part: one per press, the same for every magnet size. X = along the die table, +Z = front.
2. Laser-mark the drawing number on a side face.
3. Rev C: spring-loaded sleeve that carries the upper size ring. Guided by the plug through the ring; floats on 4 shoulder screws.
4. Faces parallel 0.01. Pocket walls square to the faces within 0.01.
5. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
6. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 31 / 33
TITLE: **UPPER CARRIER (UNIVERSAL)**			
DRAWING NO.: **MP-210**	QTY: 1 per press	SCALE: 1:1	UNITS: mm
MATERIAL: 42CrMo4 (1.7225) plate, QT 28–32 HRC			
HEAT TREATMENT / FINISH: Black oxide; pocket walls ground or wire-EDM, Ra 0.8			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED

PLAN (TOP) — front at bottom



SECTION A–A

SECTION B–B (Z = ±49.25)

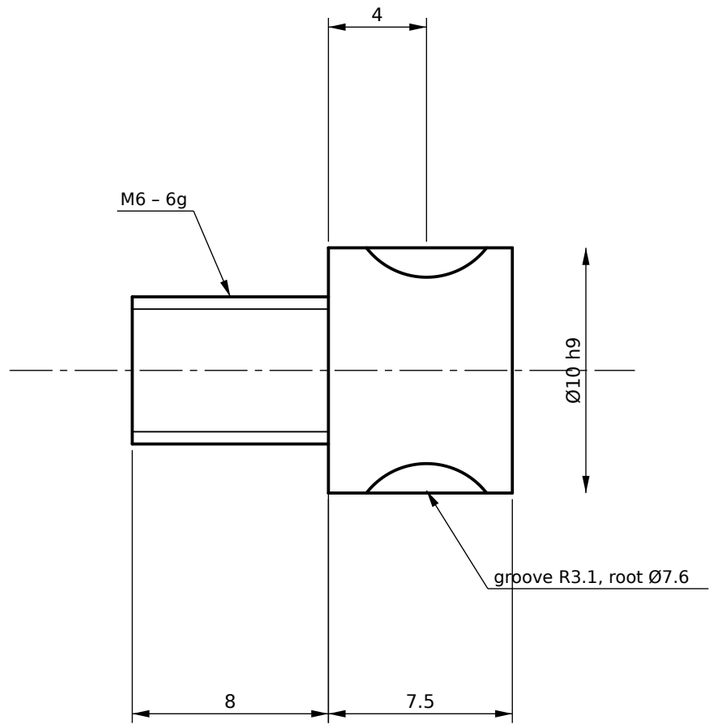
FEATURES (positions from the part centre, +Z = front)

KEY	POSITIONS (X, Z)	FEATURE
W	centre	Pocket 106.02 × 81.02 (+0.01/0), R6, 15 deep from the TOP (ring seat on a 3 mm ledge)
V	centre	Ledge opening 96 × 72, R6, THRU
R	(±62.5, ±49.25)	Ø6.1 THRU, CBORE Ø10.5 × 14 from the TOP (ISO 7379 Ø6 × 12 into the base)
P	(±62.5, 0), (0, ±49.25)	Ø10.5 × 15 from the BOTTOM (4 lower springs)
KA	(+53.01, −20), (−53.01, +20)	Variant A only: Ø4 H7 × 15 on the pocket wall, Ø4 m6 × 14 pressed flush (key pins)
KB	(+53.01, +20), (−53.01, −20)	Variant B only: as KA at these positions

NOTES

1. Universal die part: one per press, the same for every magnet size. X = along the die table, +Z = front.
2. Laser-mark the drawing number on a side face.
3. Rev C: one carrier per station. Variant A (pickup, left station, green band) and variant B (crimp, right station, red band) differ only in the key pins.
4. The keys stop a pickup ring going into the crimp station and the other way round. Faces parallel 0.01; ledge face flat 0.01.
5. Lower springs: 4 × ISO 10243 Ø10 × 32 green per carrier in 30 mm cavities: 2 mm preload (≈ 75 N), ≈ 0.38 kN at the 8 mm stop (31%).
6. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
7. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press			REVISION: **C.1**	SHEET: 32 / 33
TITLE: **LOWER CARRIER A / B (UNIVERSAL)**				
DRAWING NO.: **MP-211**	QTY: A + B per press	SCALE: 1:1	UNITS: mm	
MATERIAL: 42CrMo4 (1.7225) plate, QT 28–32 HRC				
HEAT TREATMENT / FINISH: Black oxide; pocket walls ground or wire-EDM, Ra 0.8				
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN:	CHECKED:



SIDE VIEW

NOTES

1. Screwed into the plug MP-203 (M6) with Loctite 243 until the shoulder seats. It only holds the plug up: the dowels locate it.
2. The Ø6 ball-lock pin in the body runs tangent to the groove; any rotation of the stud works.
3. Groove: R3.1 centred 4.0 above the shoulder, root Ø7.6. Top end chamfer 0.5 × 45°.
4. General tolerances ISO 2768-mK; fits per ISO 286; threads ISO metric 6H/6g.
5. Break all sharp edges 0.3 × 45° unless stated. Remove all burrs.

PROJECT: MagPress M4 · manual photo-magnet press		REVISION: **C.1**	SHEET: 33 / 33
TITLE: **PLUG STUD (SIZE KIT)**			
DRAWING NO.: **MP-212**	QTY: 1 per size kit (4)	SCALE: 4:1	UNITS: mm
MATERIAL: C45 Ø12 bar, QT 28–32 HRC			
HEAT TREATMENT / FINISH: Black oxide			
GENERAL TOLERANCES: ISO 2768-mK	1st	DATE: 2026-10-27	DRAWN / CHECKED